

BFS Traversal

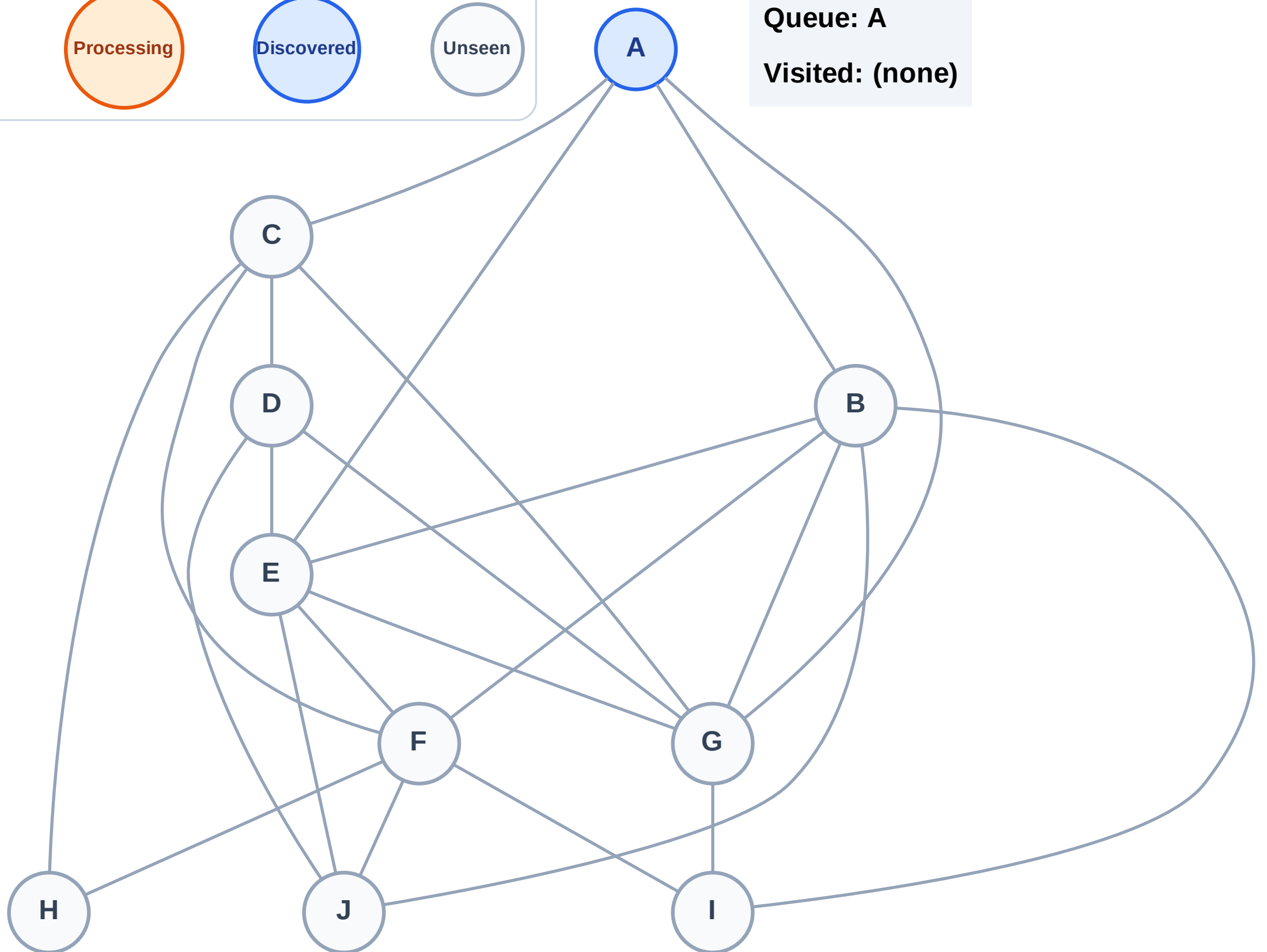
Step 1: Start at A

Legend



Queue: A

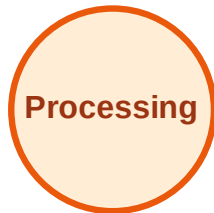
Visited: (none)



BFS Traversal

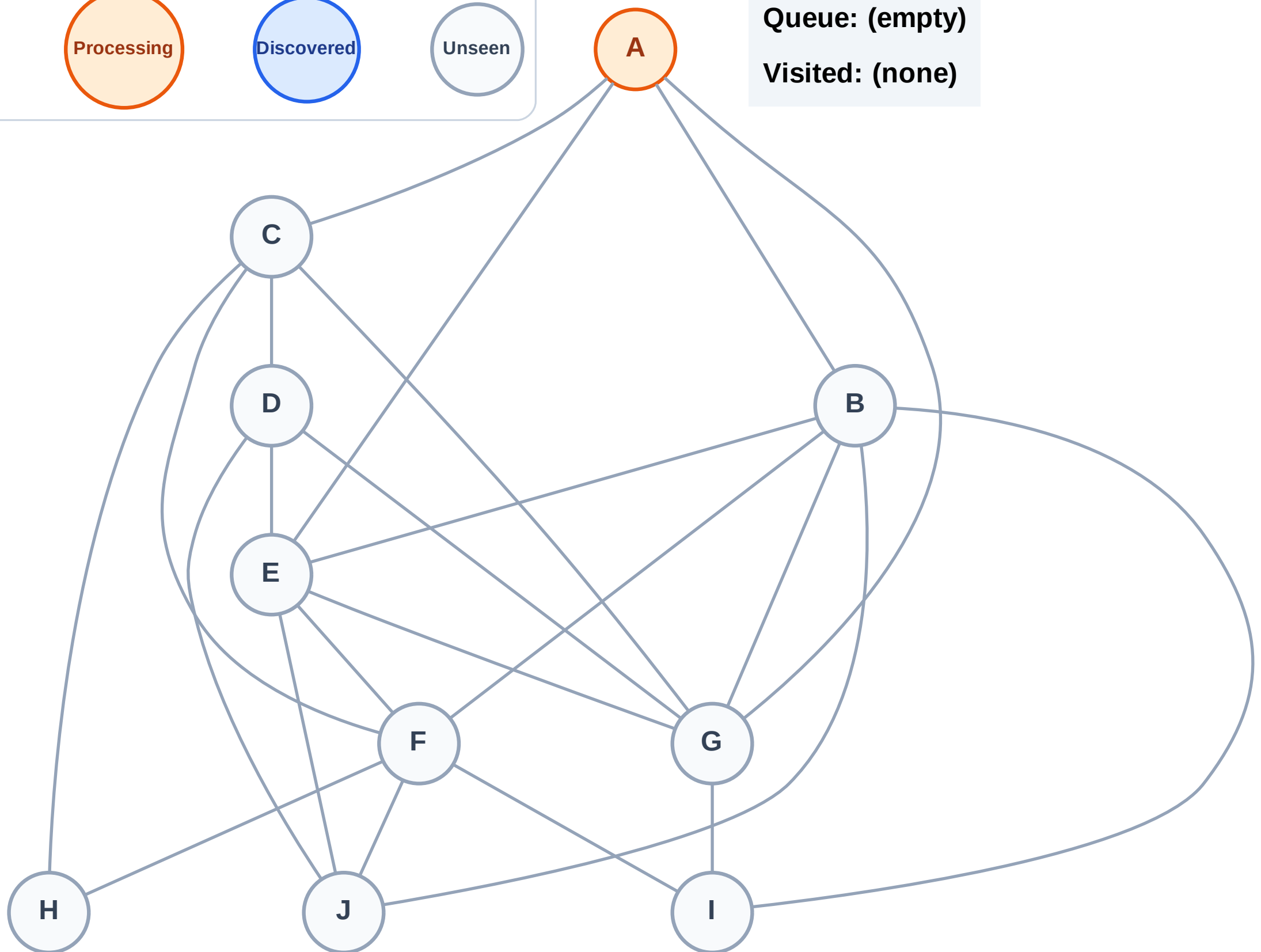
Step 2: Process node A

Legend



Queue: (empty)

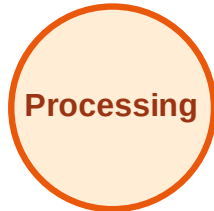
Visited: (none)



BFS Traversal

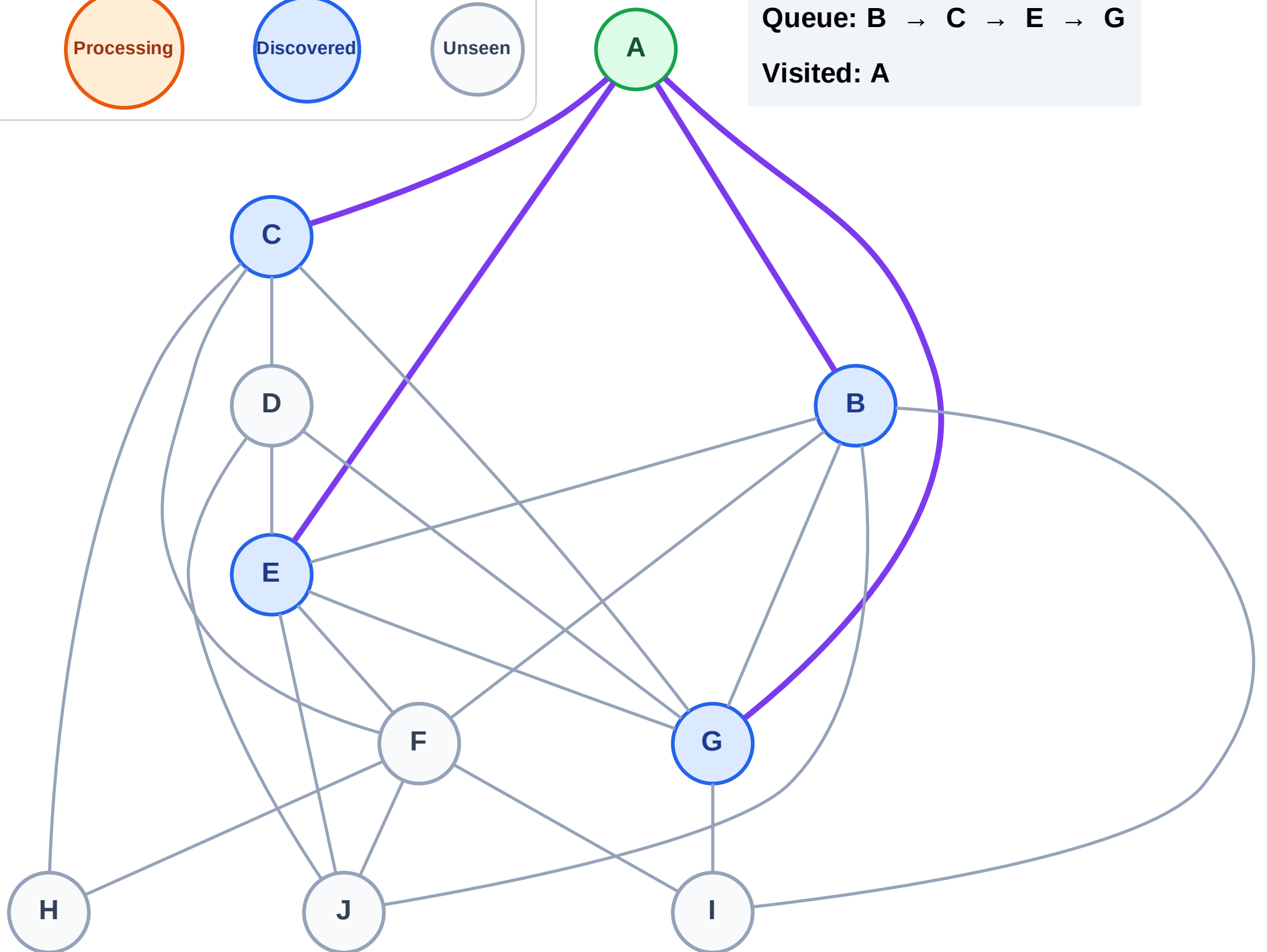
Step 3: Discover B, C, E, G from A

Legend



Queue: B → C → E → G

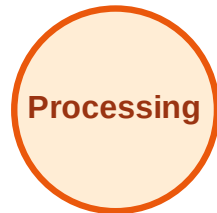
Visited: A



BFS Traversal

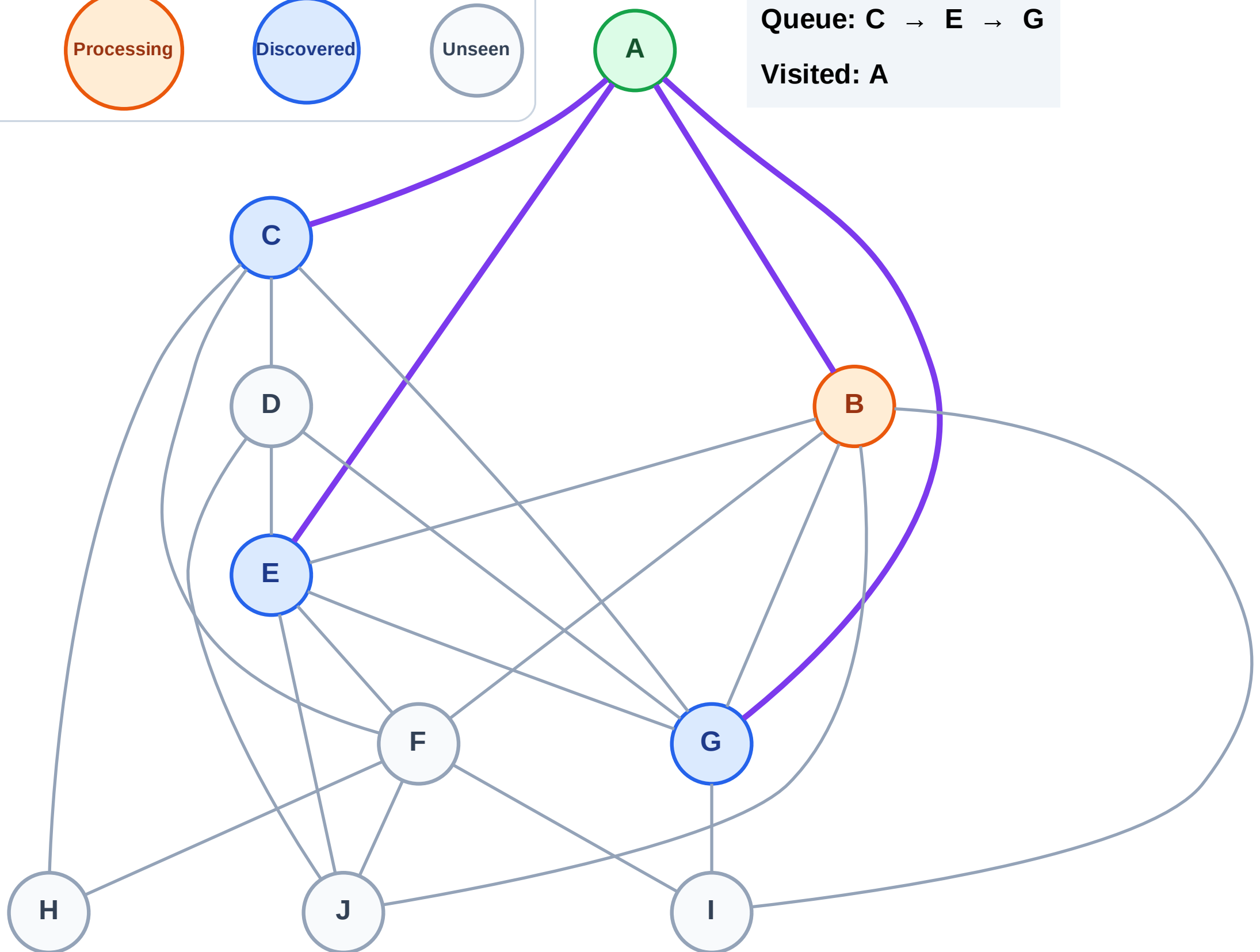
Step 4: Process node B

Legend



Queue: C → E → G

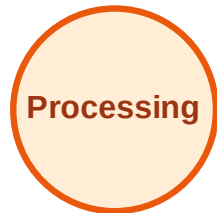
Visited: A



BFS Traversal

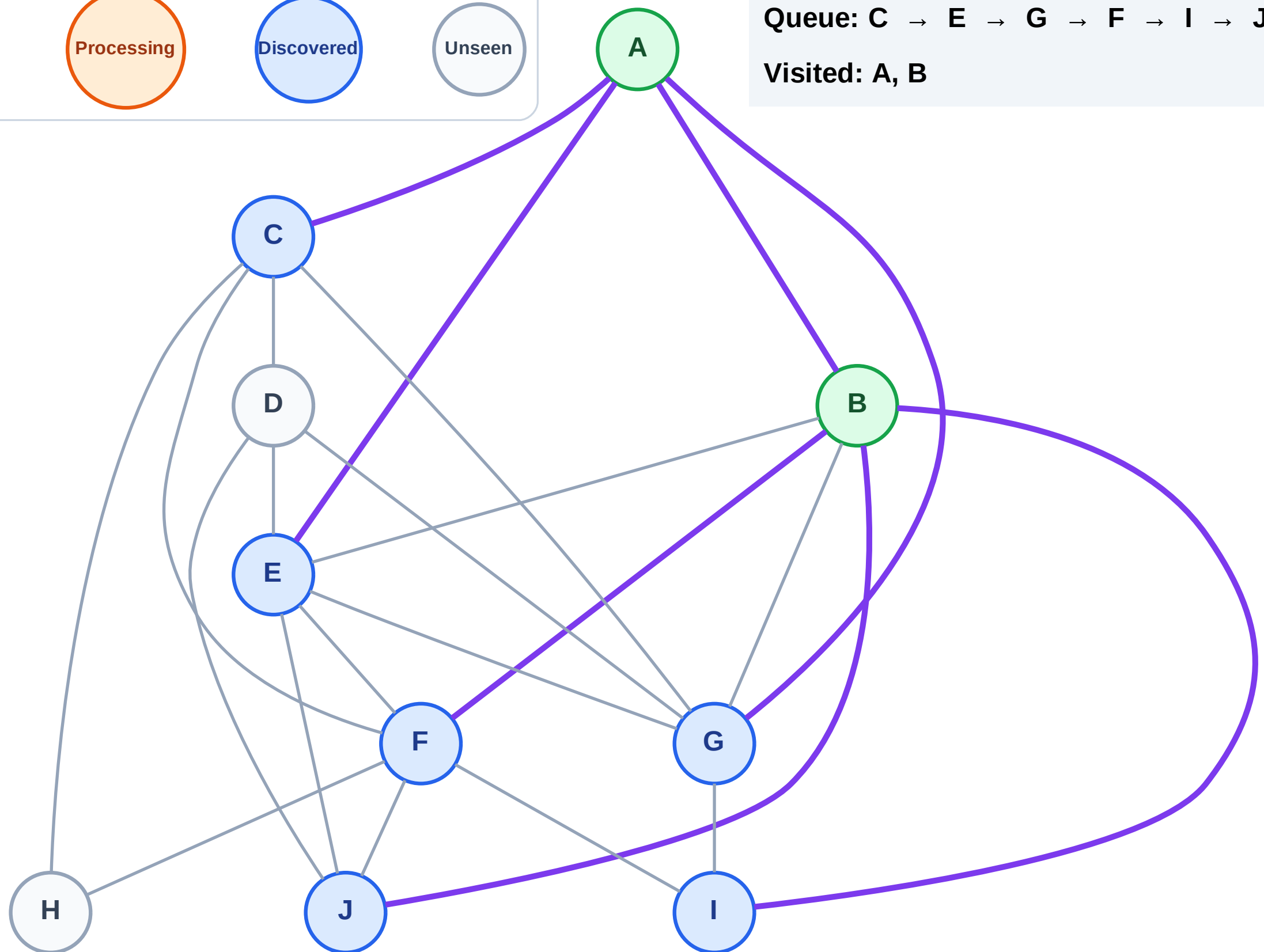
Step 5: Discover F, I, J from B

Legend



Queue: C → E → G → F → I → J

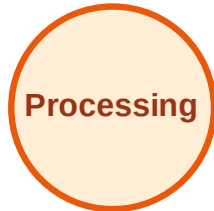
Visited: A, B



BFS Traversal

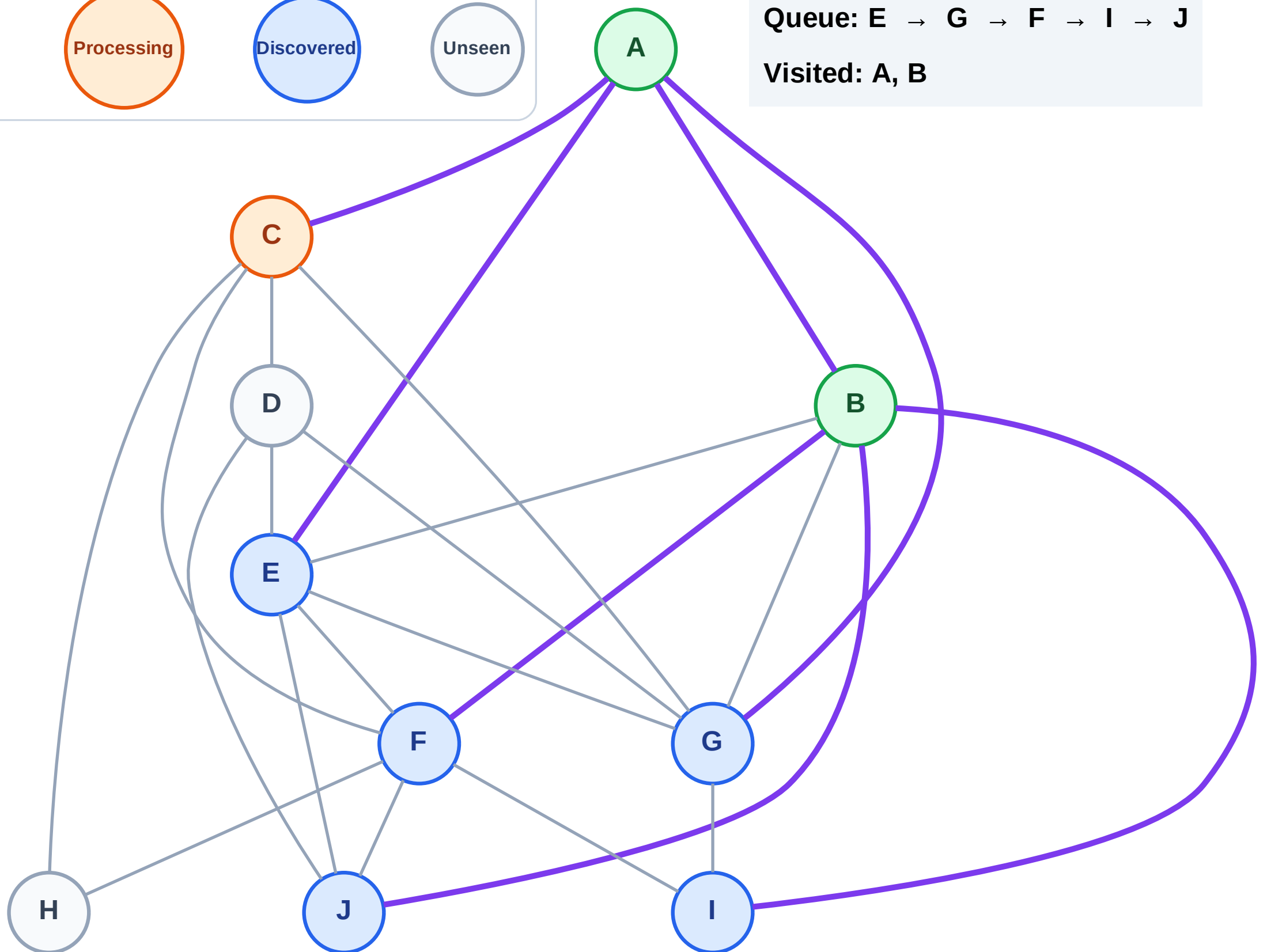
Step 6: Process node C

Legend



Queue: E → G → F → I → J

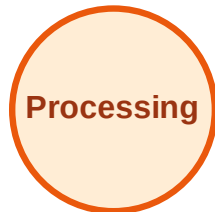
Visited: A, B



BFS Traversal

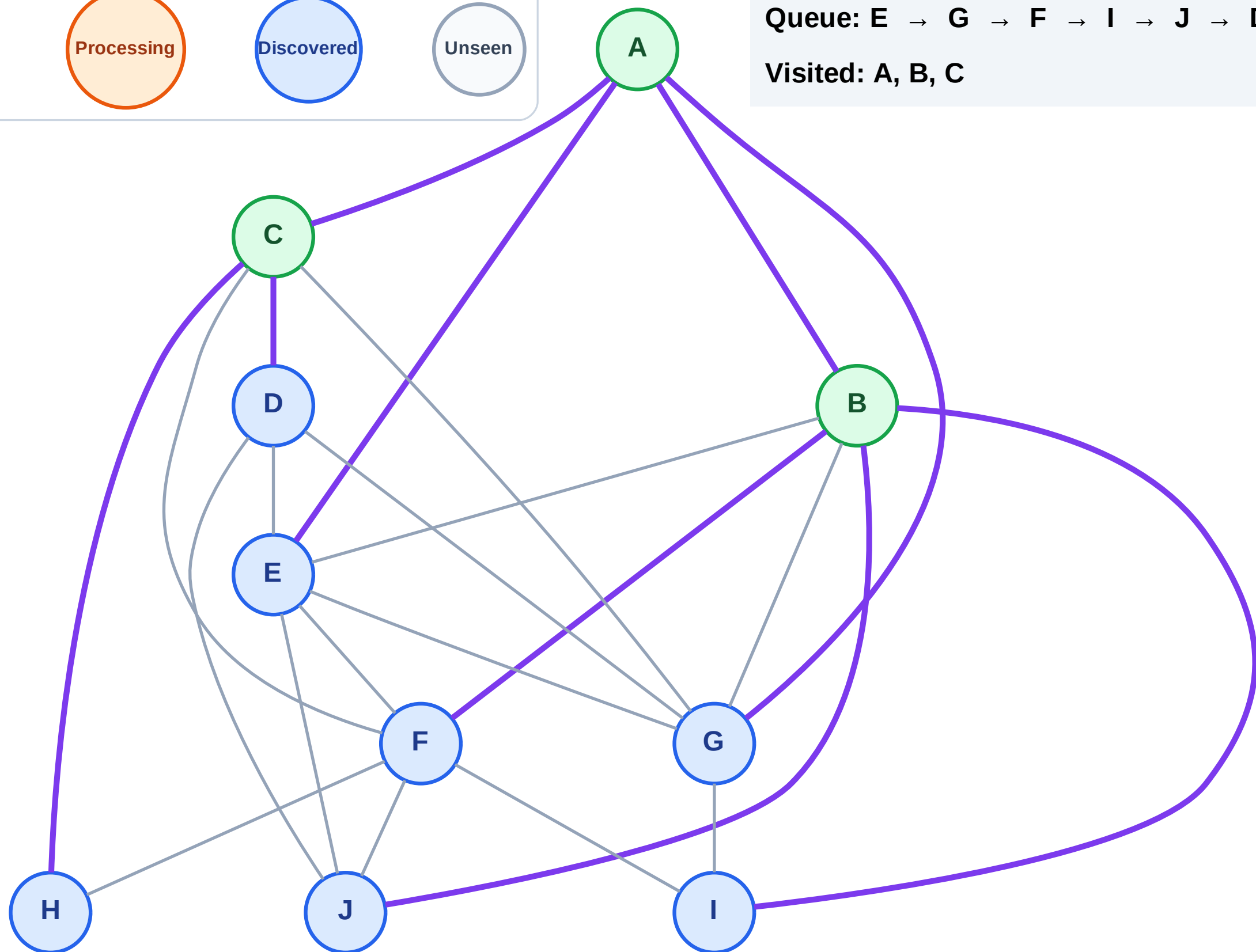
Step 7: Discover D, H from C

Legend



Queue: E → G → F → I → J → D → H

Visited: A, B, C



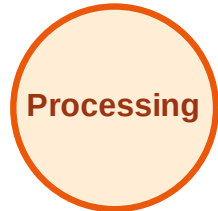
BFS Traversal

Step 8: Process node E

Legend



Complete



Processing



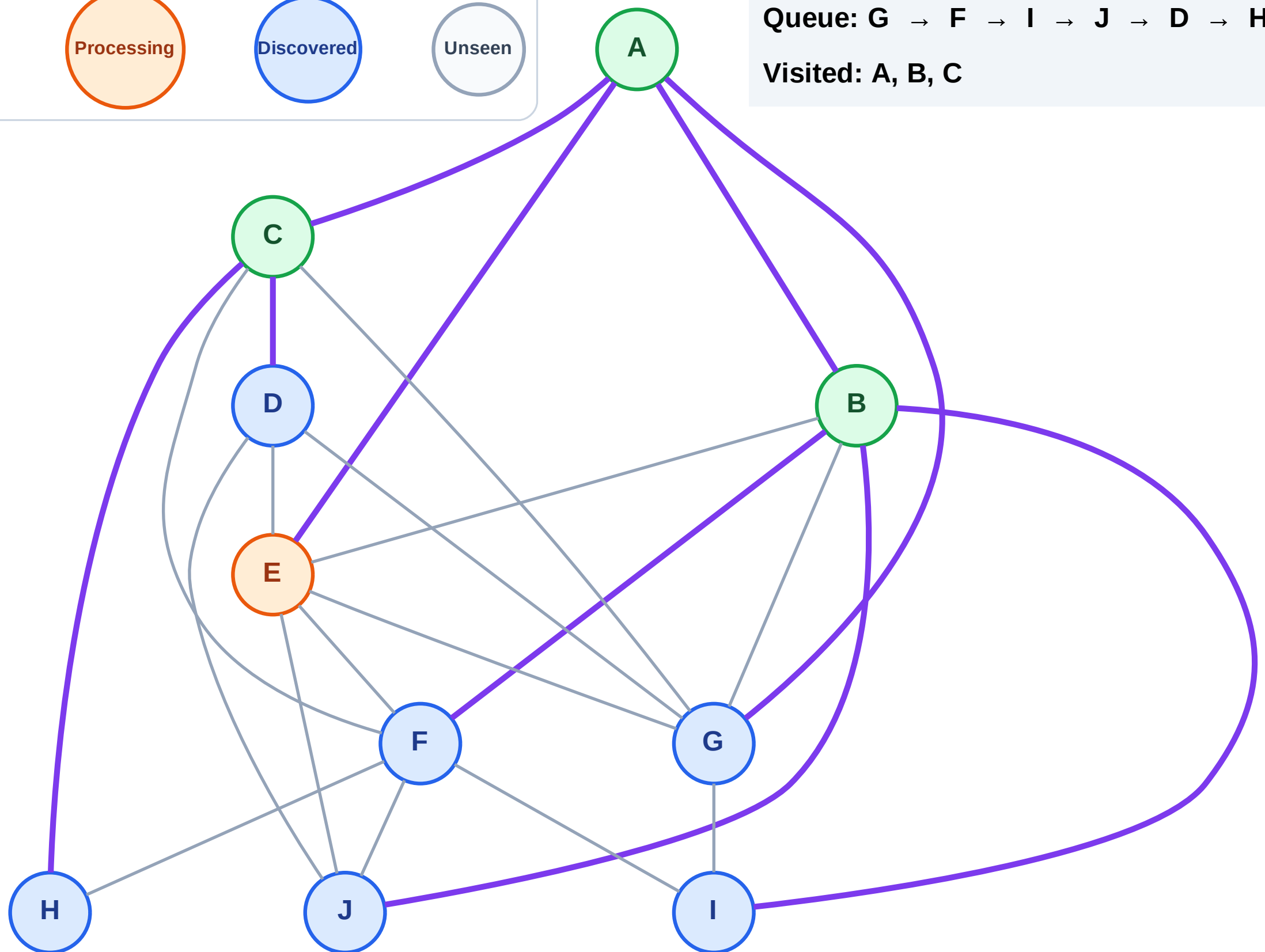
Discovered



Unseen

Queue: G → F → I → J → D → H

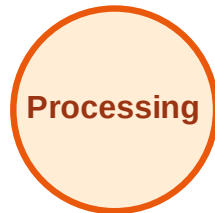
Visited: A, B, C



BFS Traversal

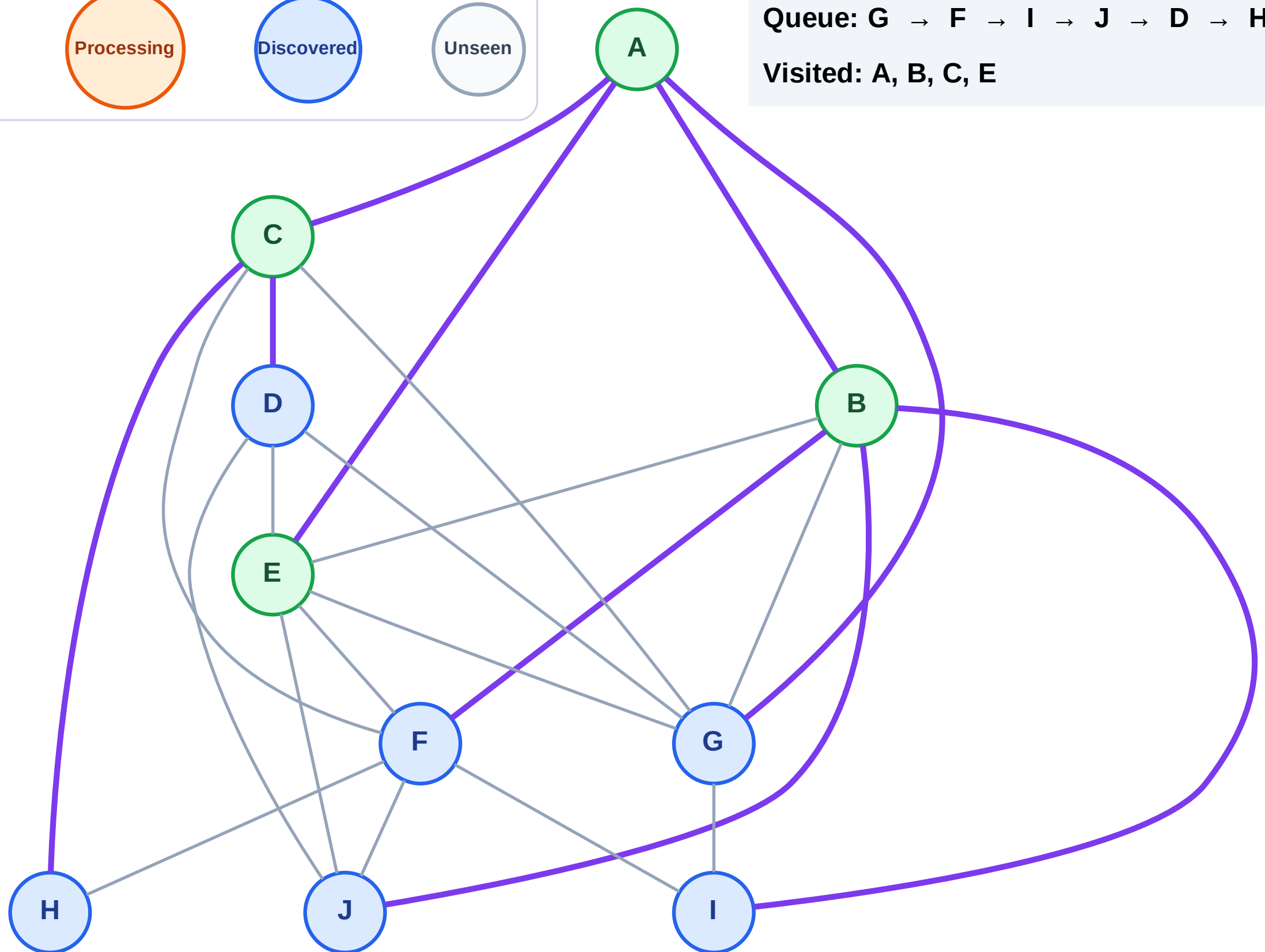
Step 9: No new neighbors from E

Legend



Queue: G → F → I → J → D → H

Visited: A, B, C, E



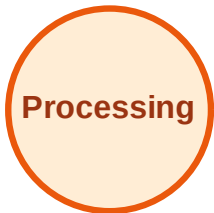
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



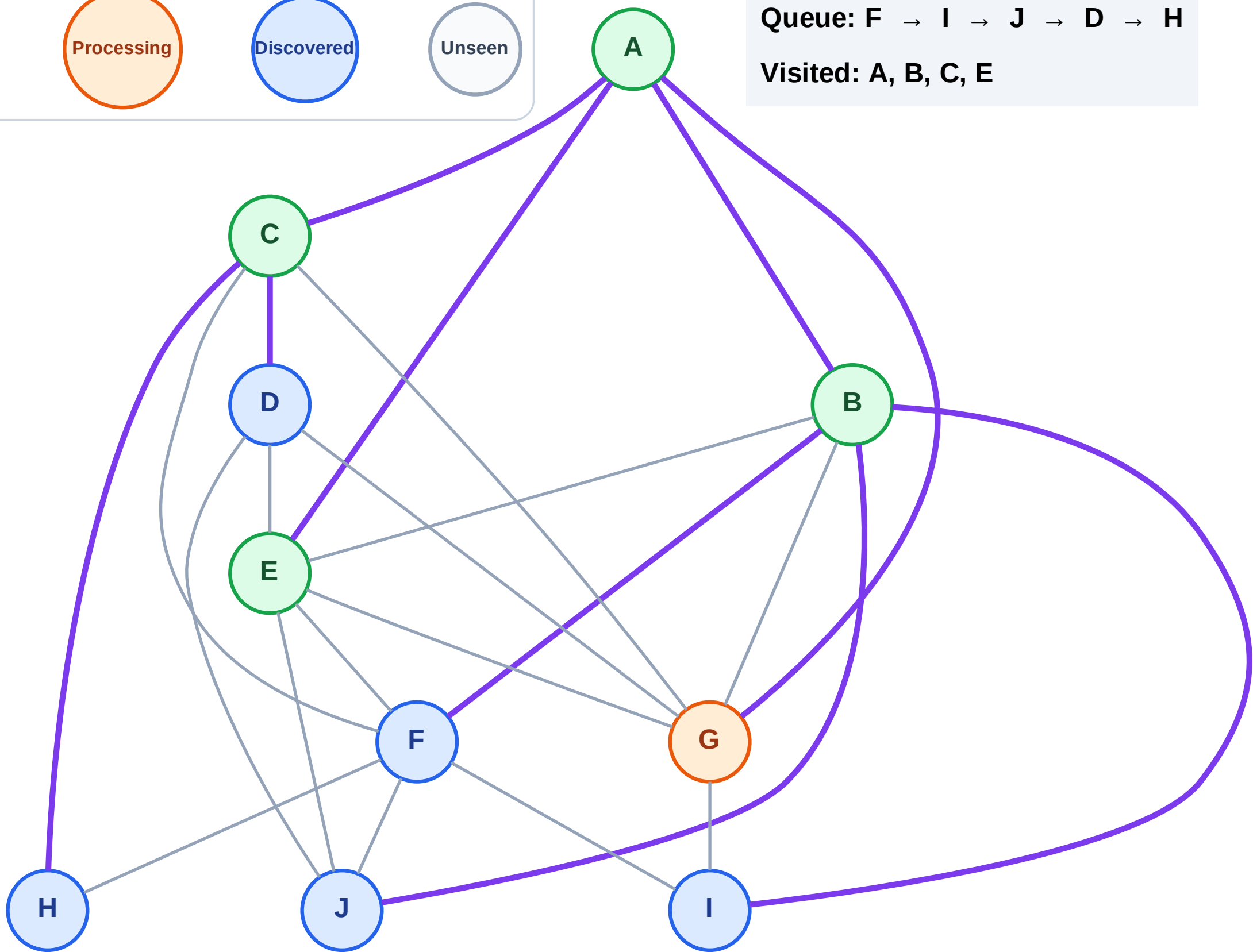
Discovered



Unseen

Queue: F → I → J → D → H

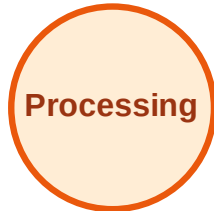
Visited: A, B, C, E



BFS Traversal

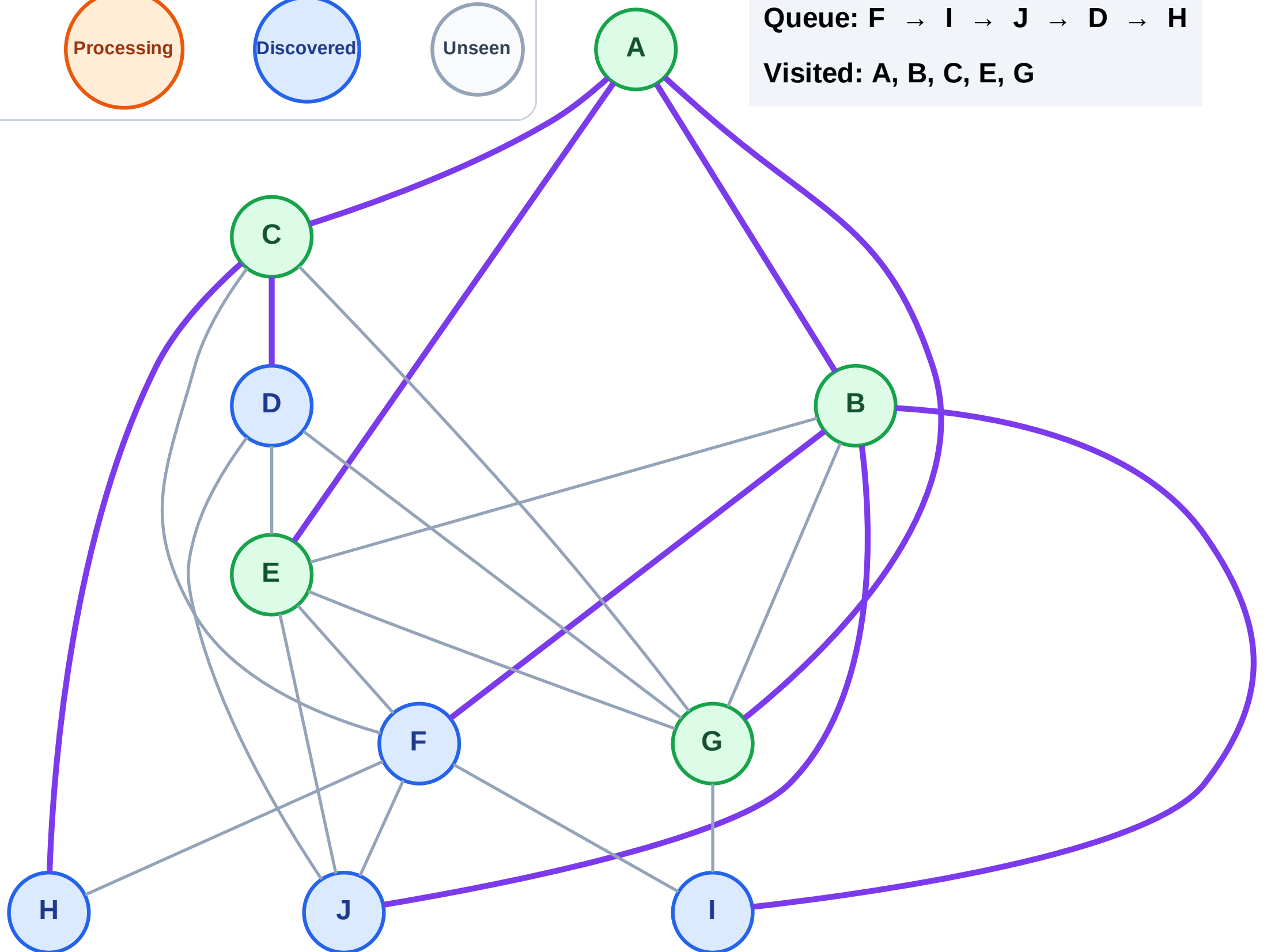
Step 11: No new neighbors from G

Legend



Queue: F → I → J → D → H

Visited: A, B, C, E, G



BFS Traversal

Step 12: Process node F

Legend

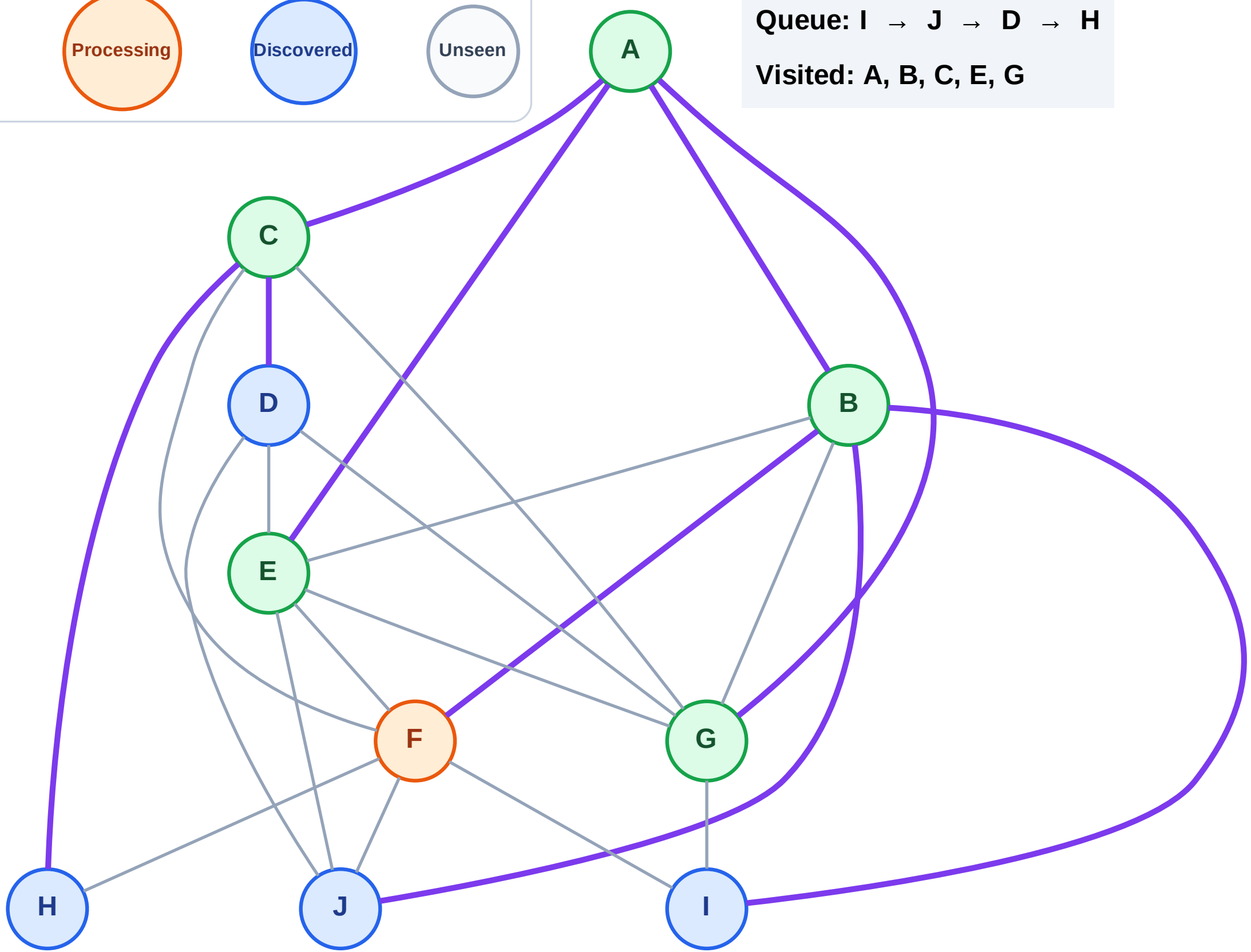
Complete

Processing

Discovered

Unseen

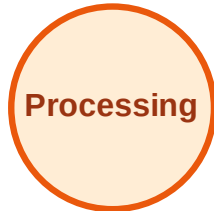
Queue: I → J → D → H
Visited: A, B, C, E, G



BFS Traversal

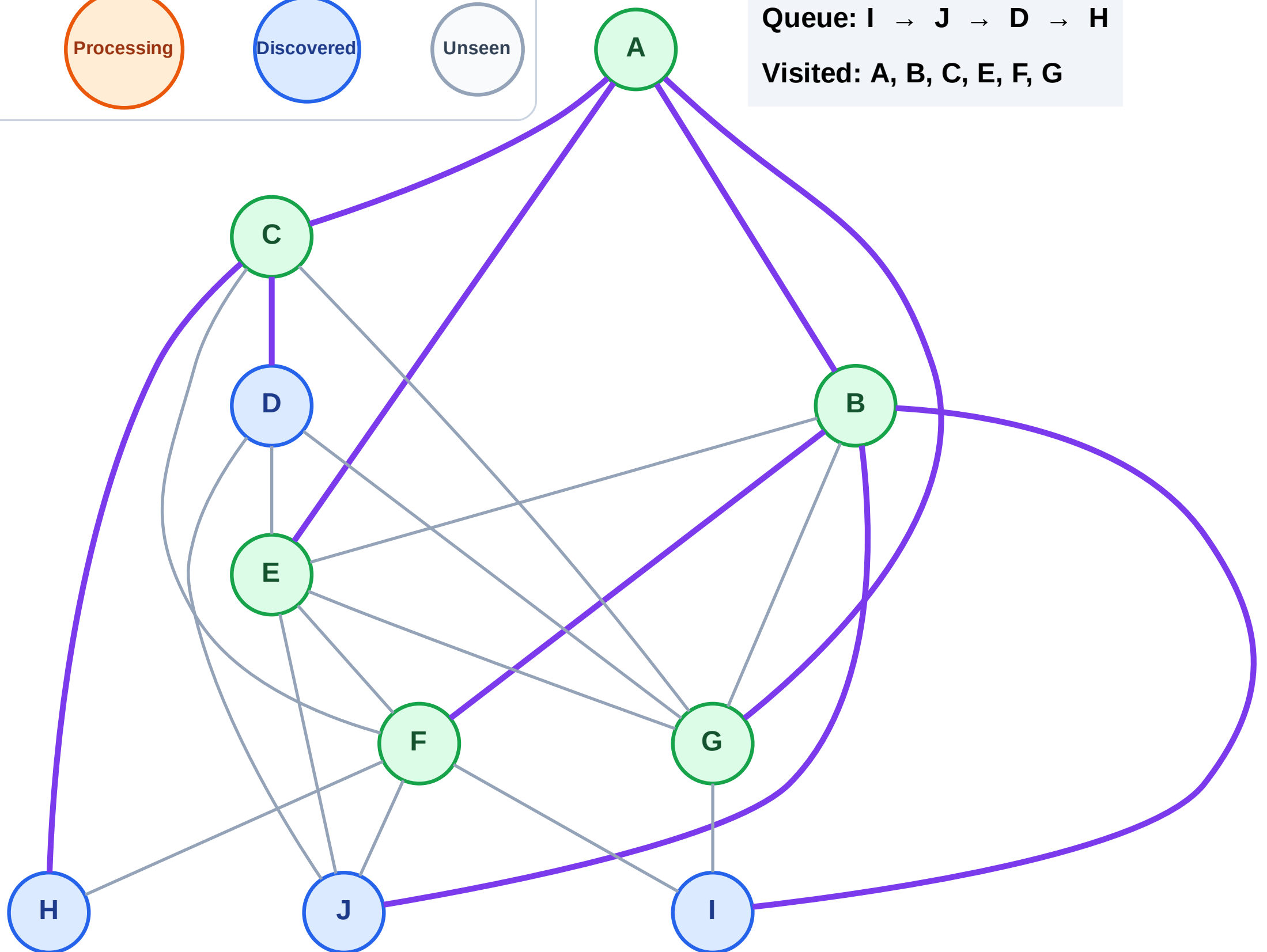
Step 13: No new neighbors from F

Legend



Queue: I → J → D → H

Visited: A, B, C, E, F, G



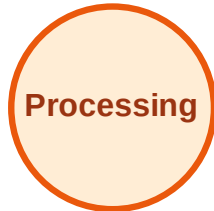
BFS Traversal

Step 14: Process node I

Legend



Complete



Processing



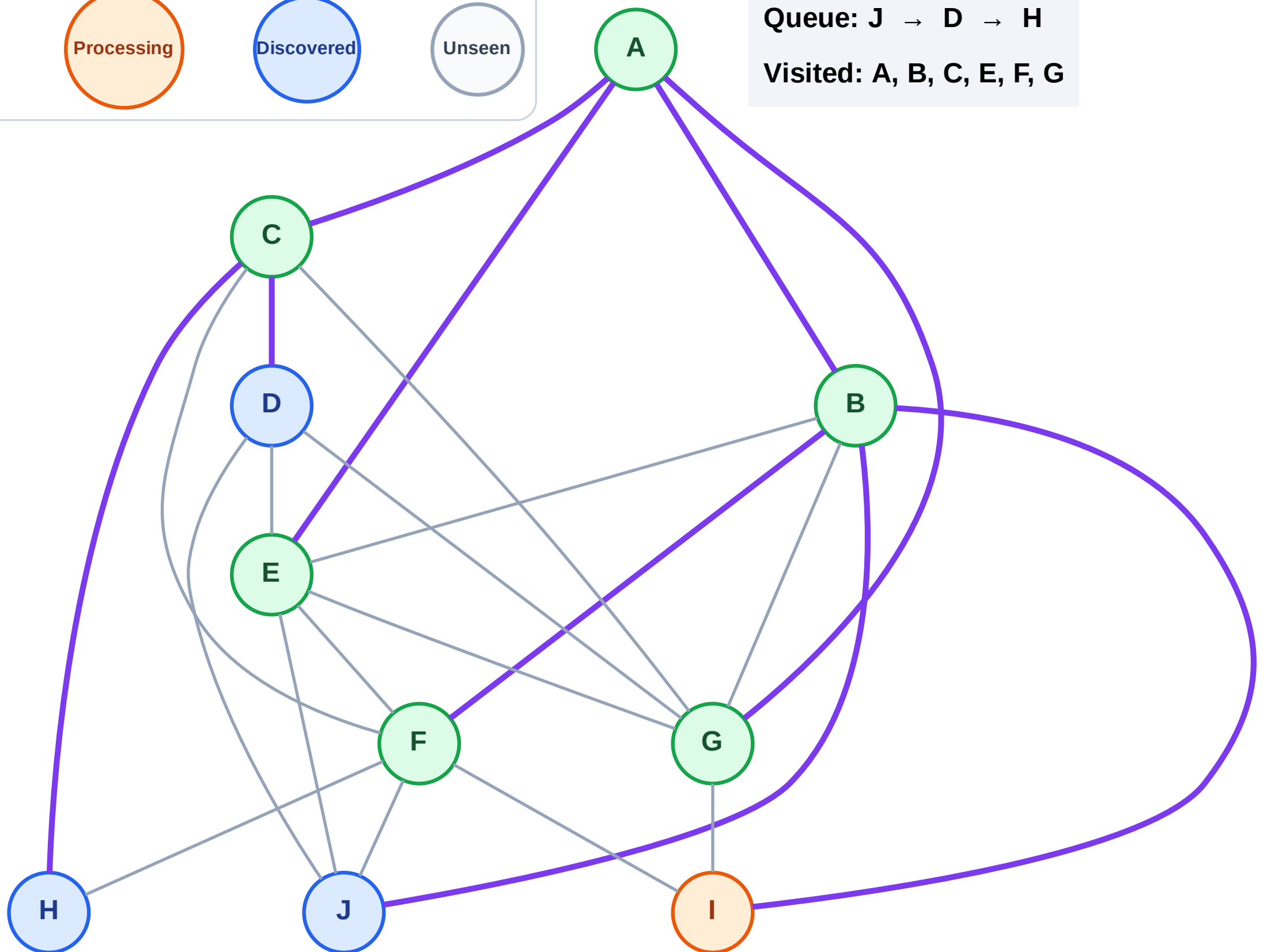
Discovered



Unseen

Queue: J → D → H

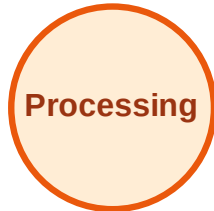
Visited: A, B, C, E, F, G



BFS Traversal

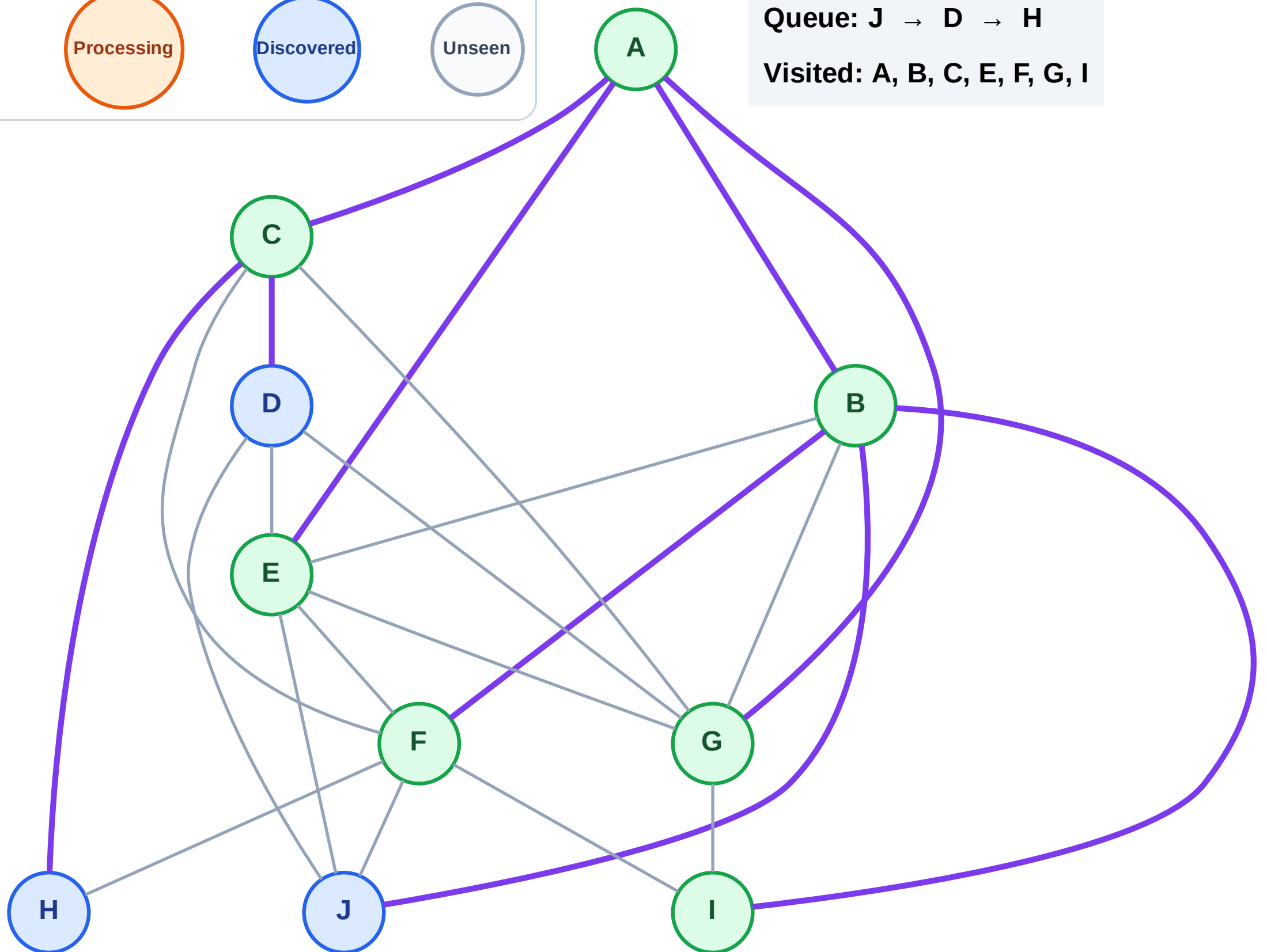
Step 15: No new neighbors from I

Legend



Queue: J → D → H

Visited: A, B, C, E, F, G, I



BFS Traversal

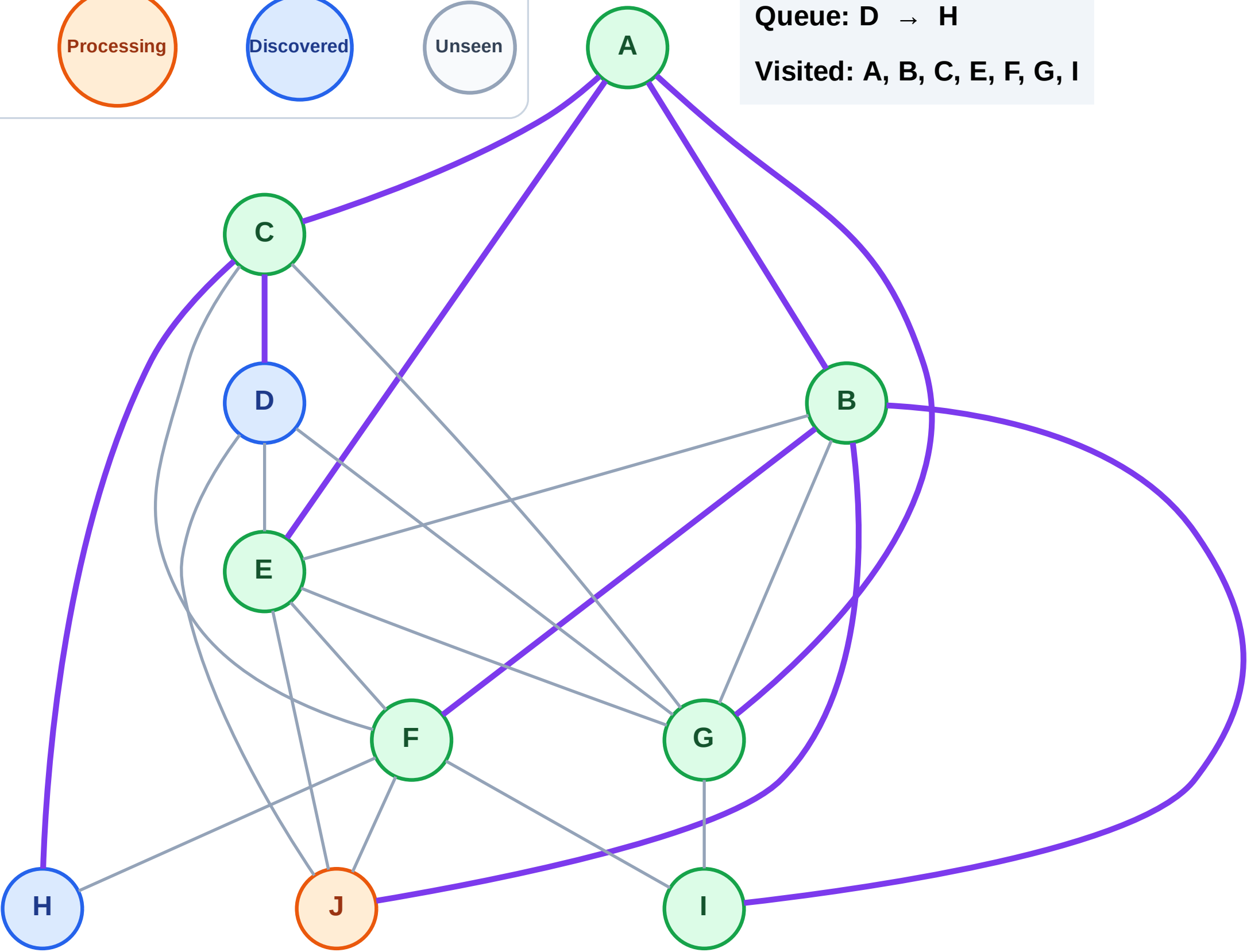
Step 16: Process node J

Legend



Queue: D → H

Visited: A, B, C, E, F, G, I



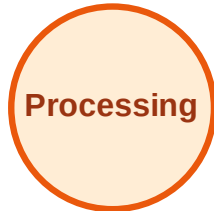
BFS Traversal

Step 17: No new neighbors from J

Legend



Complete



Processing



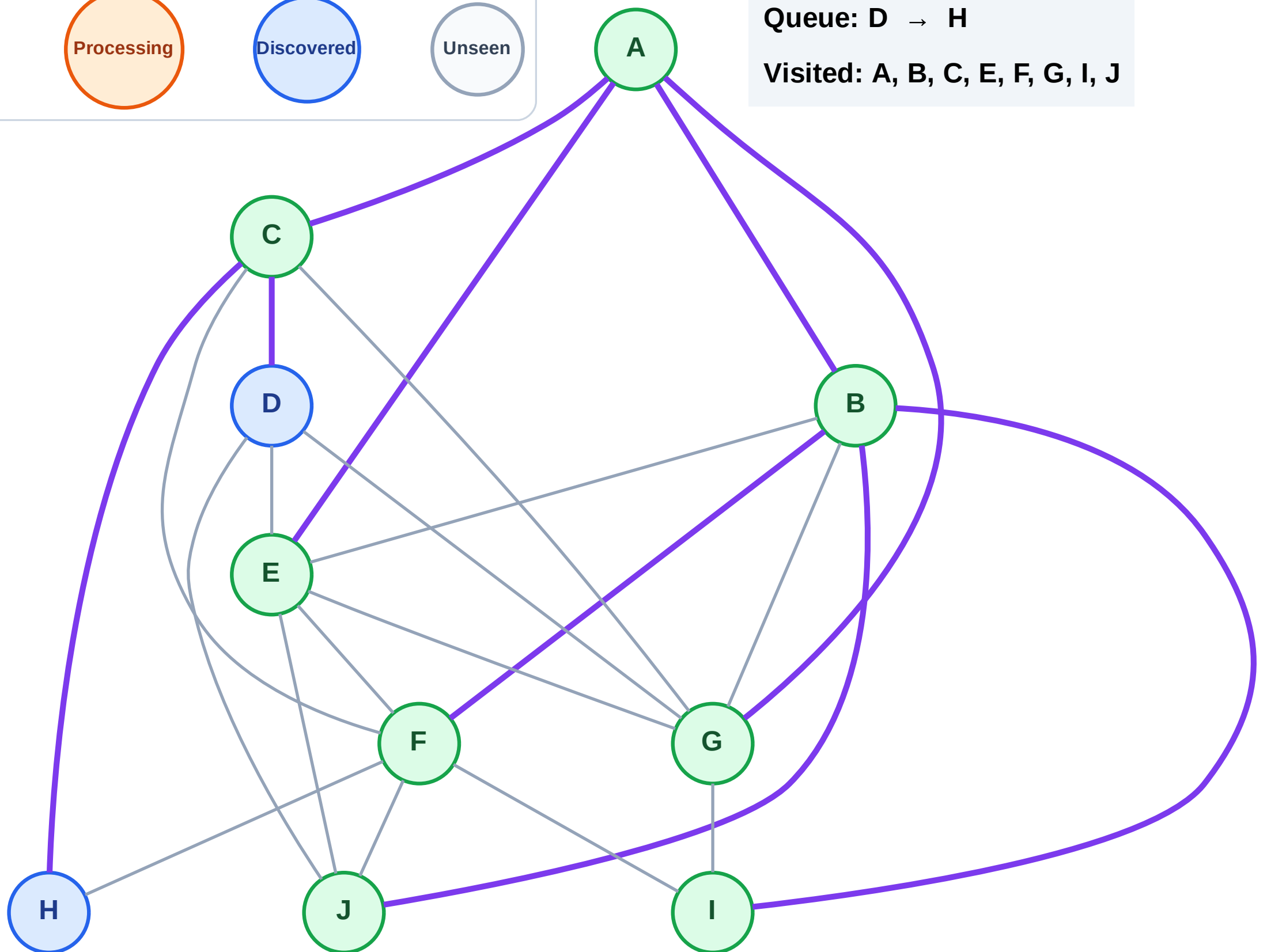
Discovered



Unseen

Queue: D → H

Visited: A, B, C, E, F, G, I, J



BFS Traversal

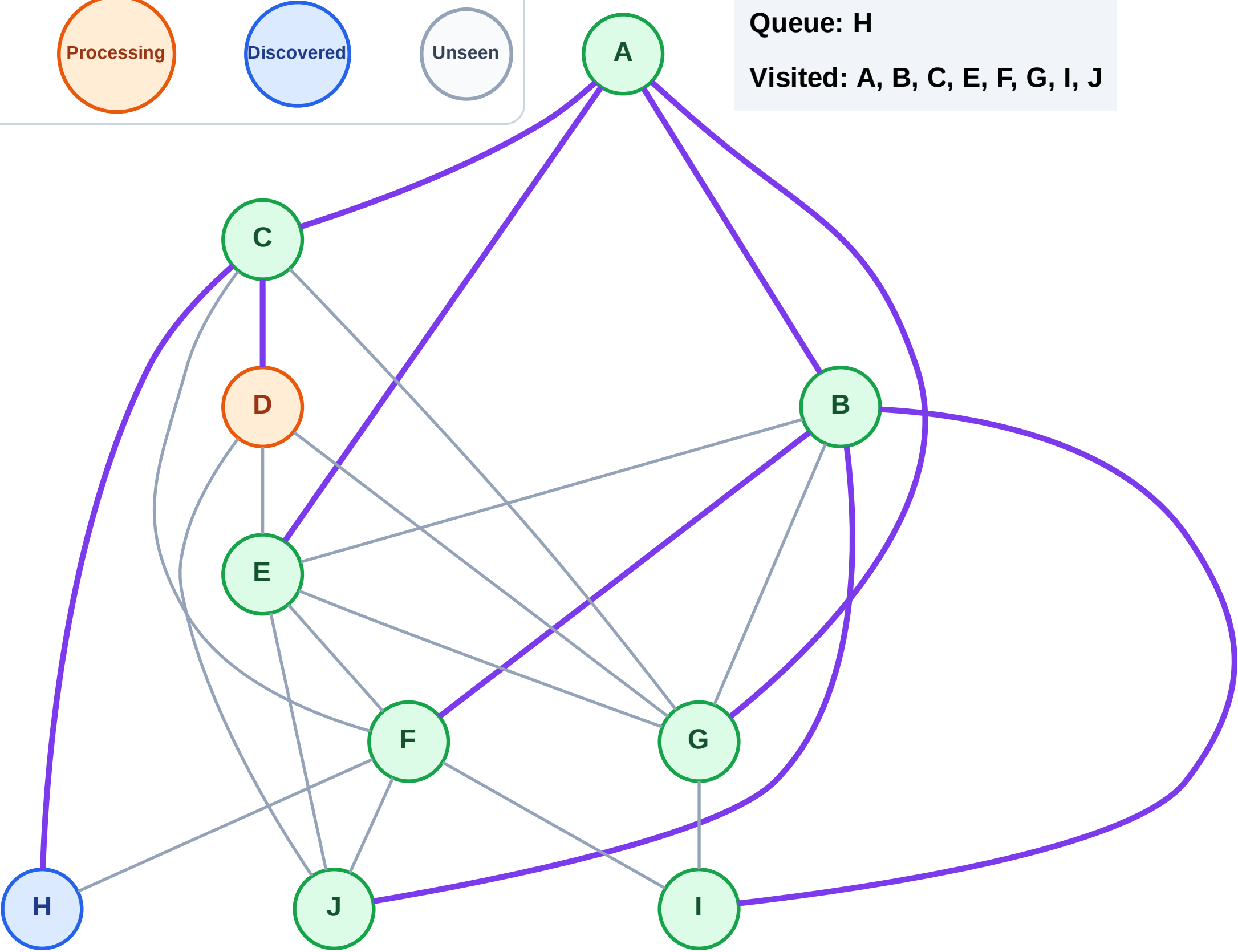
Step 18: Process node D

Legend



Queue: H

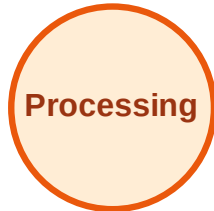
Visited: A, B, C, E, F, G, I, J



BFS Traversal

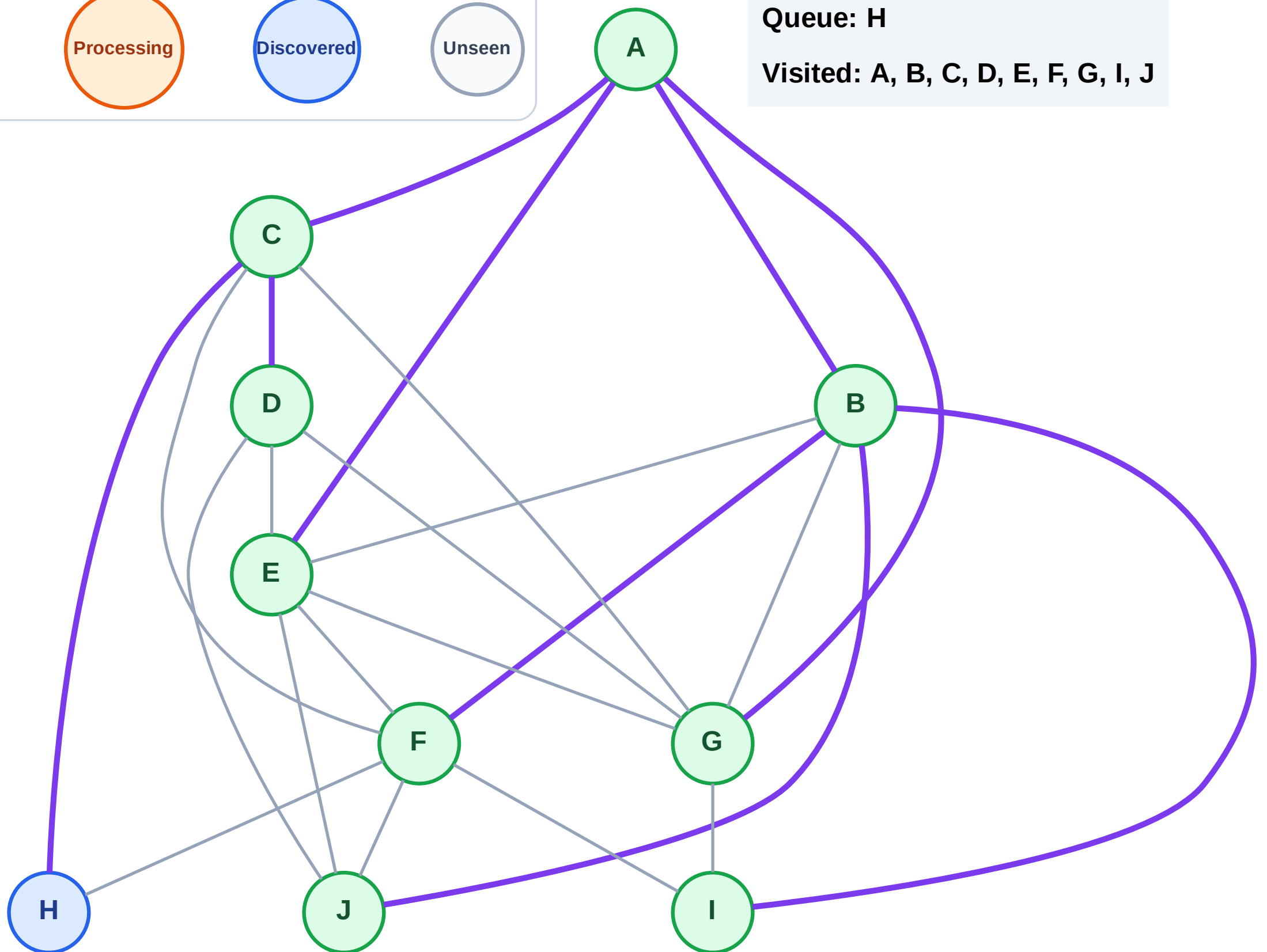
Step 19: No new neighbors from D

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

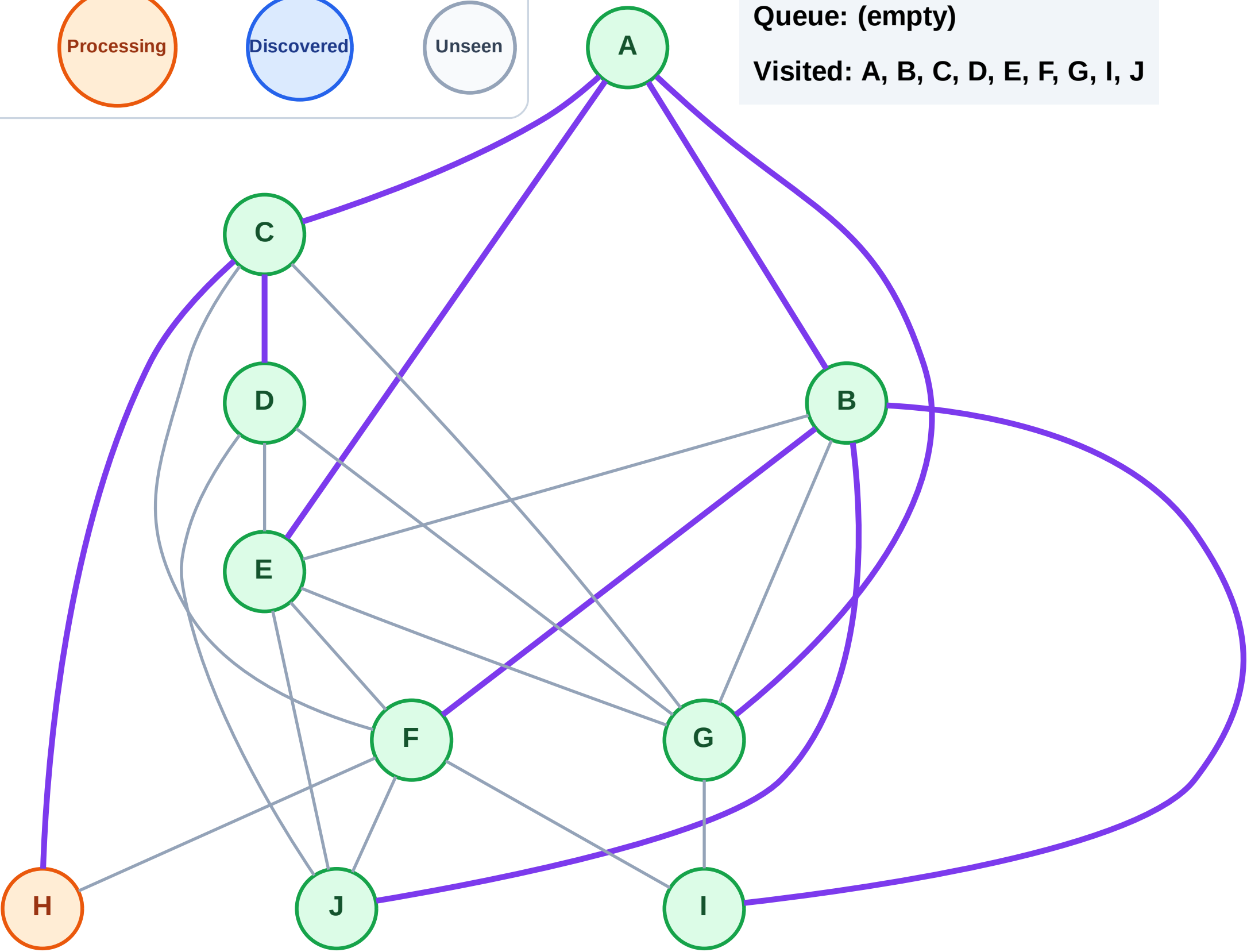
Step 20: Process node H

Legend



Queue: (empty)

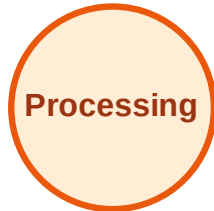
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

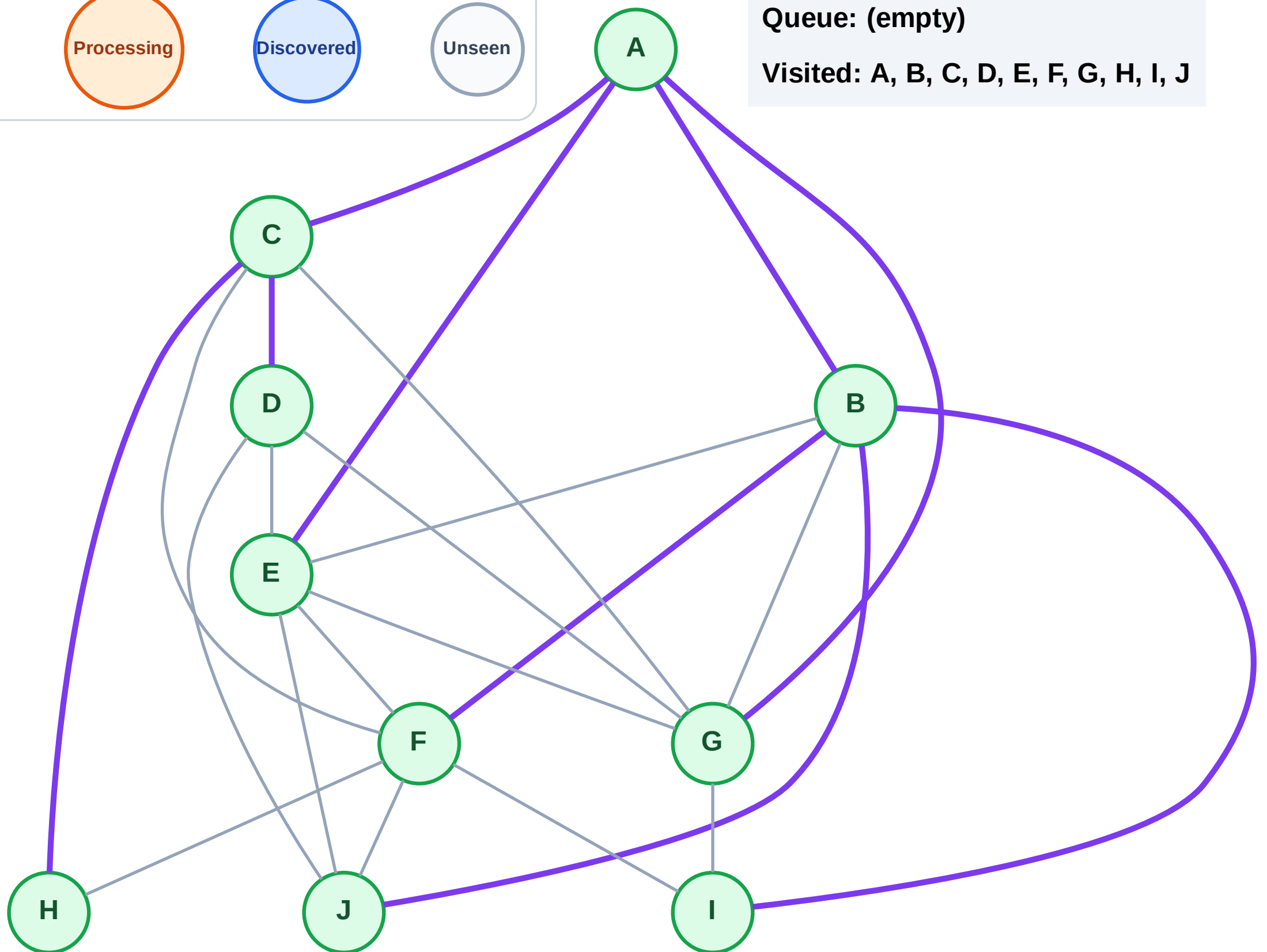
Step 21: No new neighbors from H

Legend



Queue: (empty)

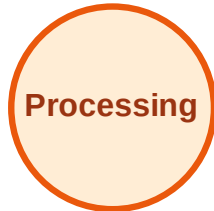
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

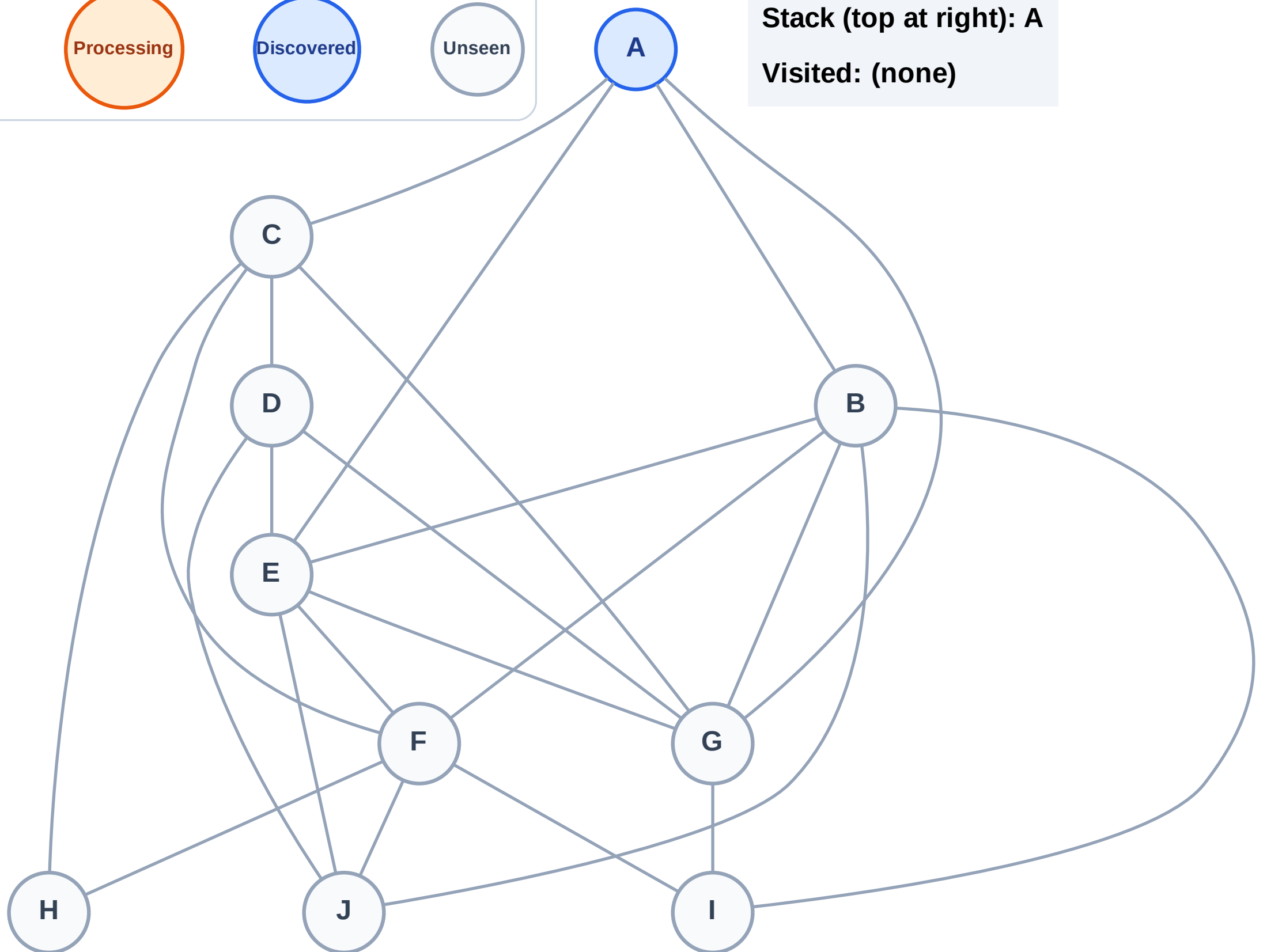
Step 1: Start at A

Legend



Stack (top at right): A

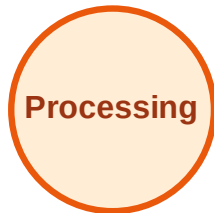
Visited: (none)



DFS Traversal

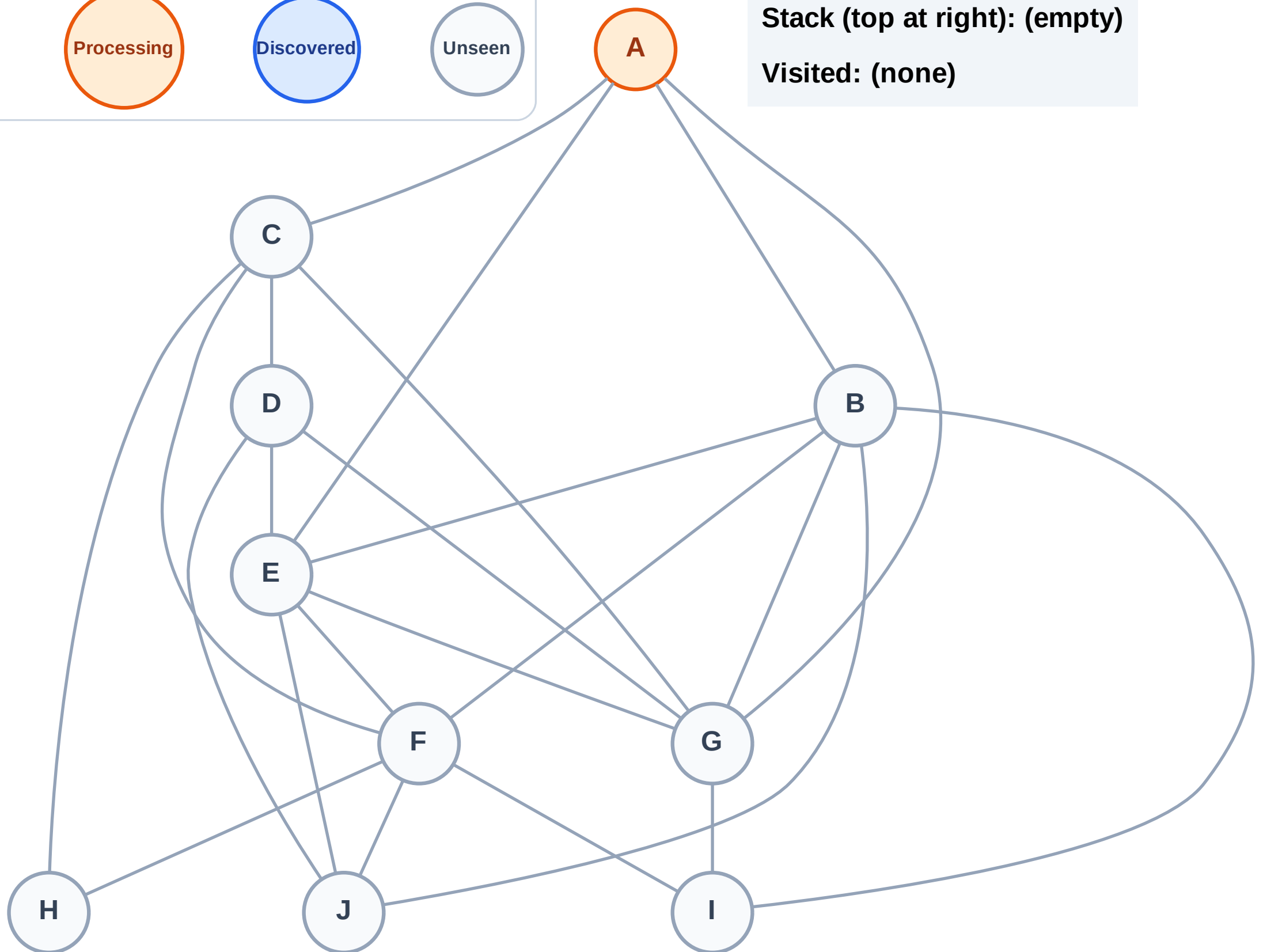
Step 2: Process node A

Legend



Stack (top at right): (empty)

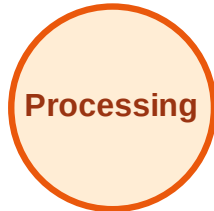
Visited: (none)



DFS Traversal

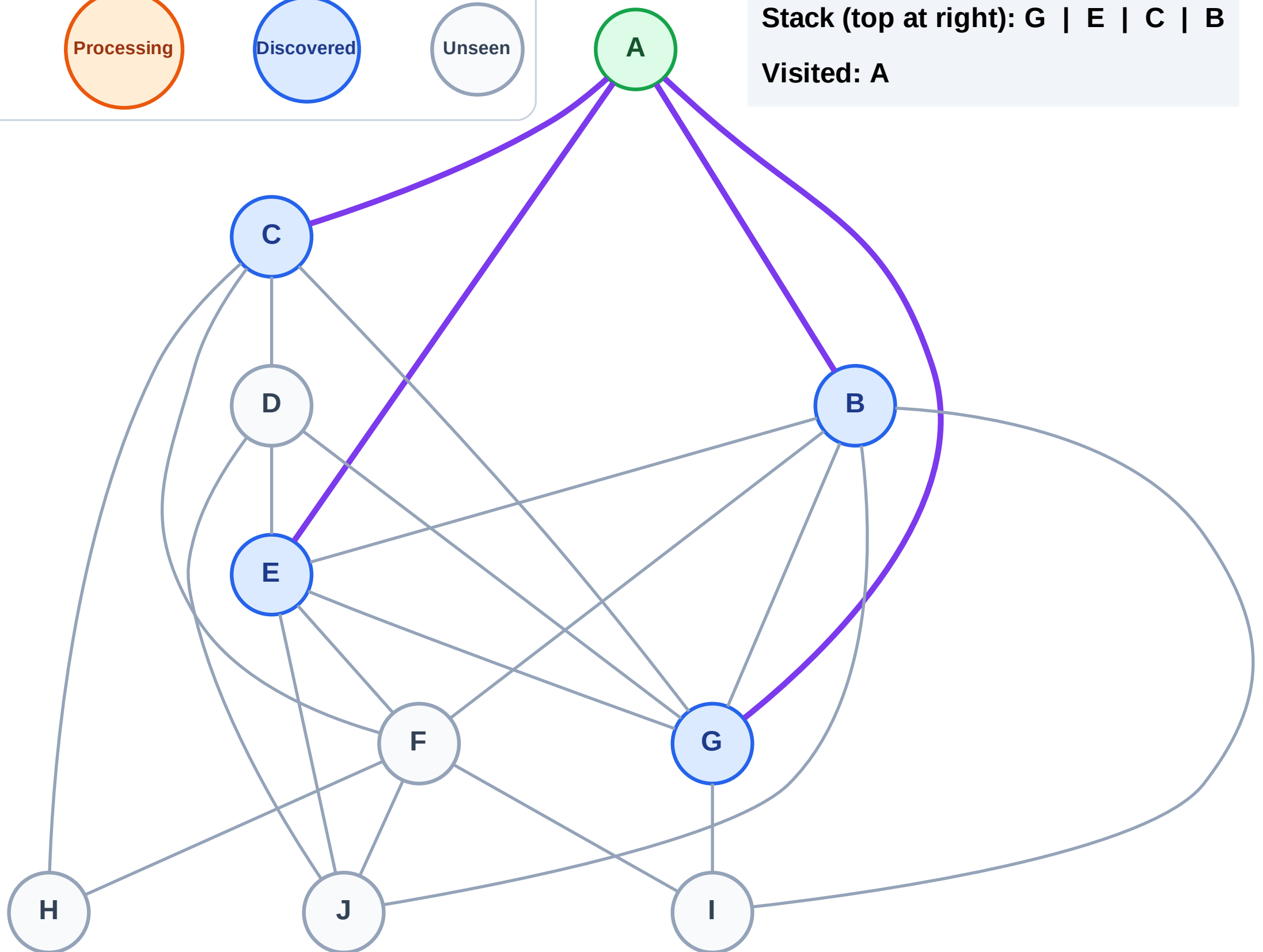
Step 3: Discover G, E, C, B from A

Legend



Stack (top at right): G | E | C | B

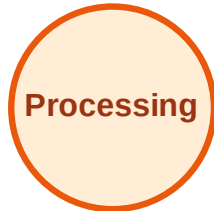
Visited: A



DFS Traversal

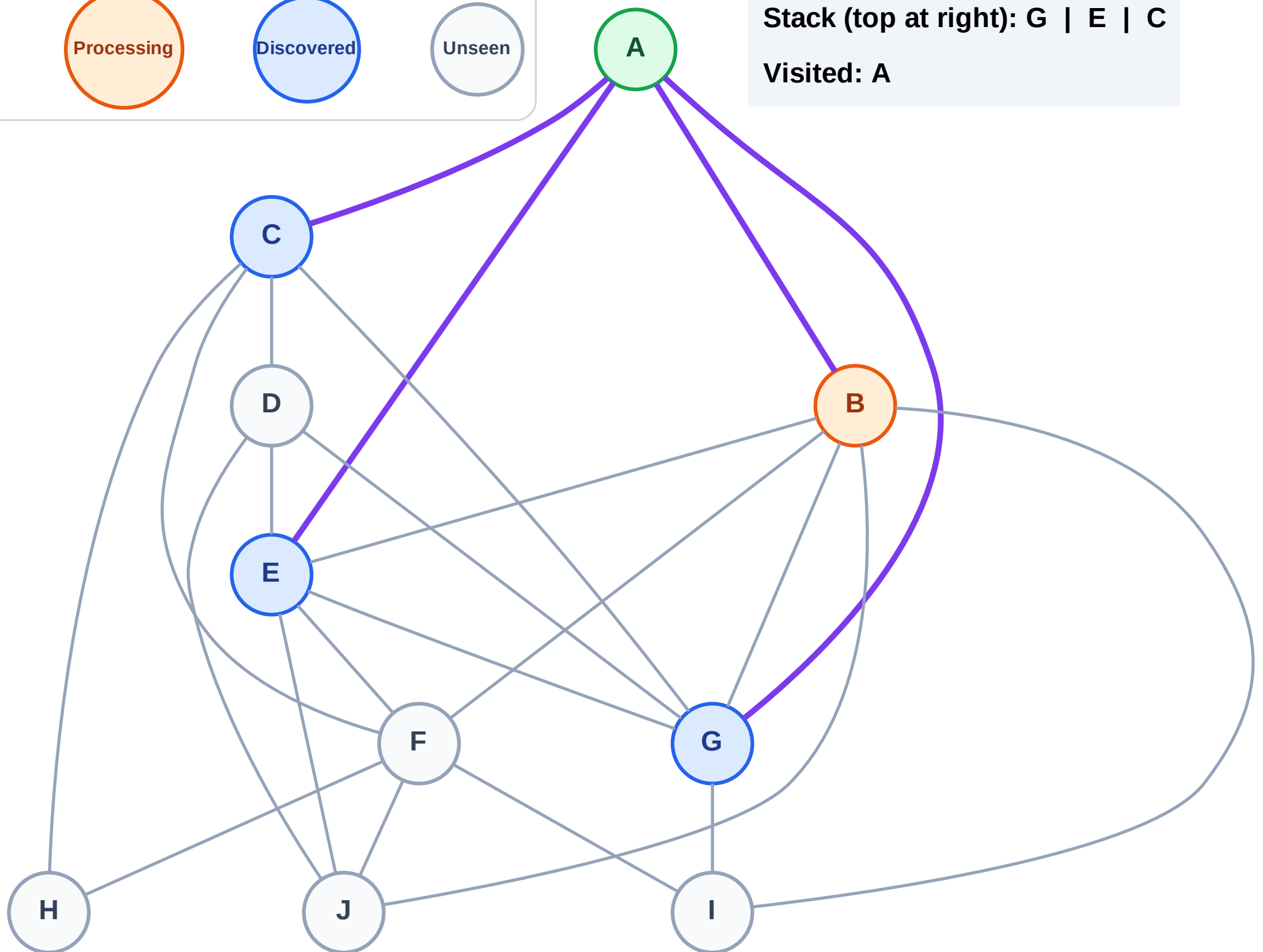
Step 4: Process node B

Legend



Stack (top at right): G | E | C

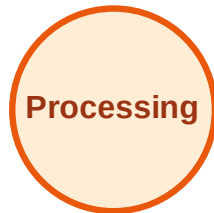
Visited: A



DFS Traversal

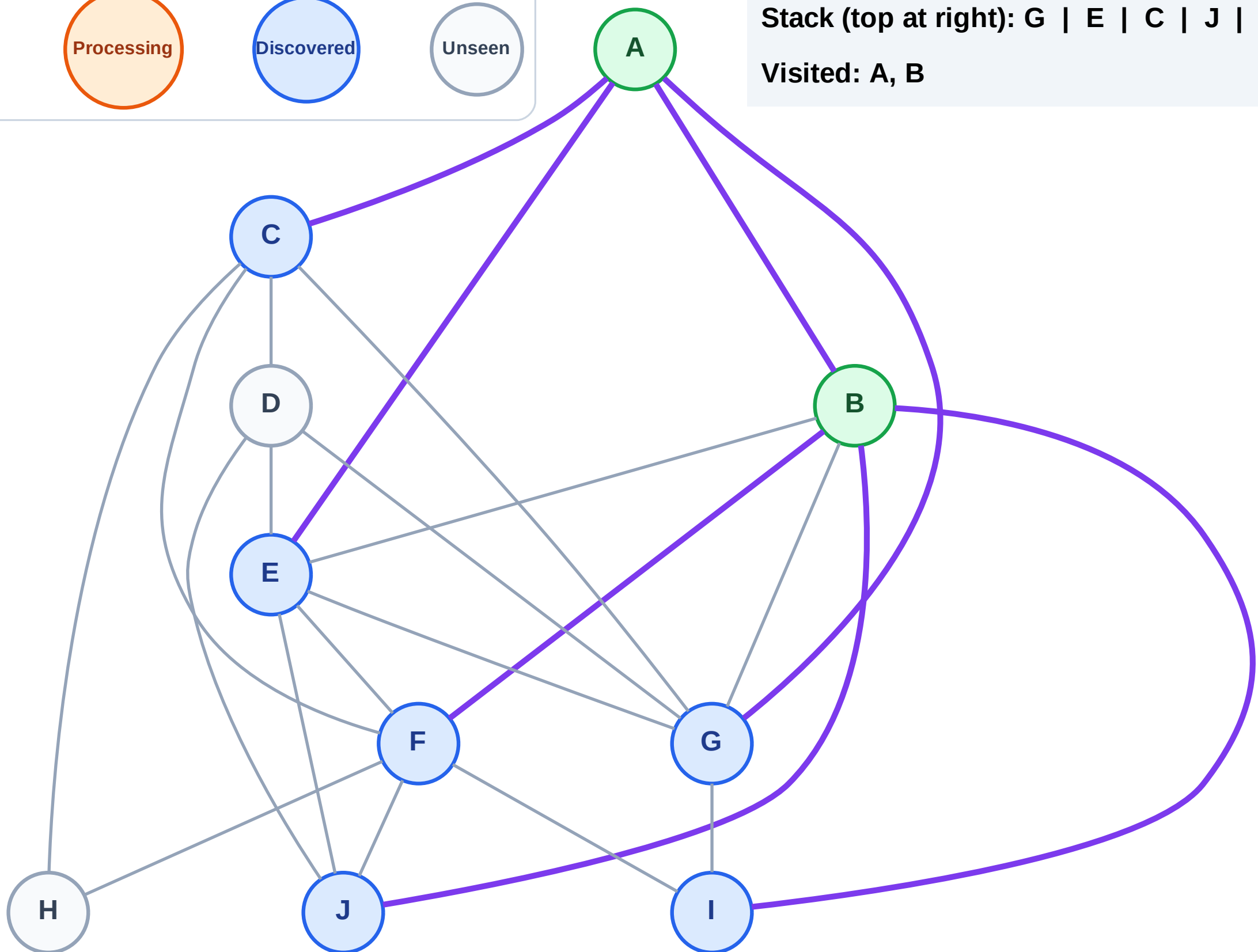
Step 5: Discover J, I, F from B

Legend



Stack (top at right): G | E | C | J | I | F

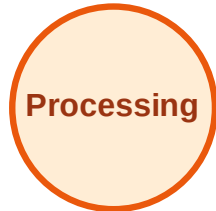
Visited: A, B



DFS Traversal

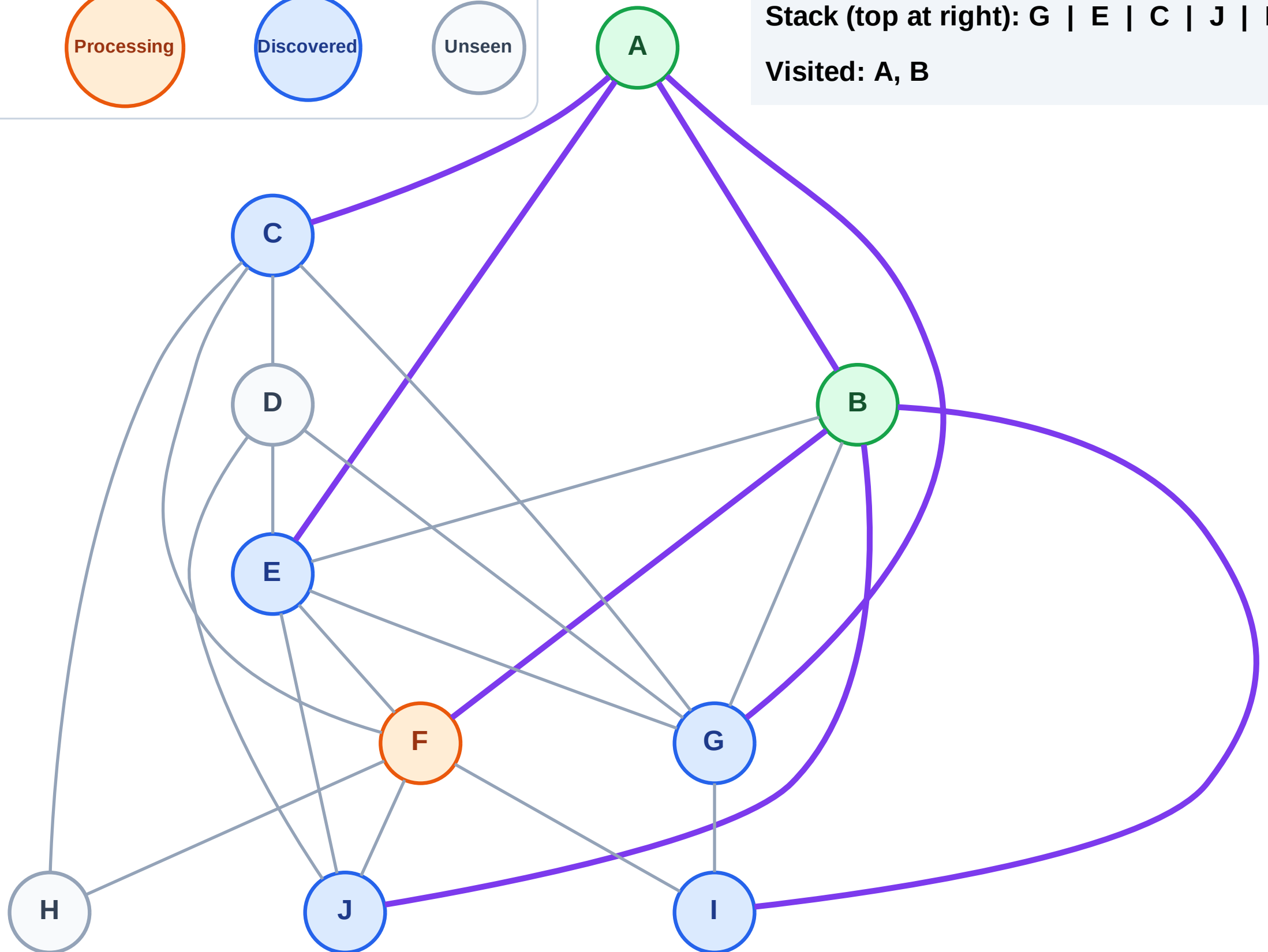
Step 6: Process node F

Legend



Stack (top at right): G | E | C | J | I

Visited: A, B



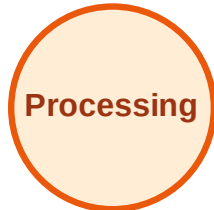
DFS Traversal

Step 7: Discover H from F

Legend



Complete



Processing



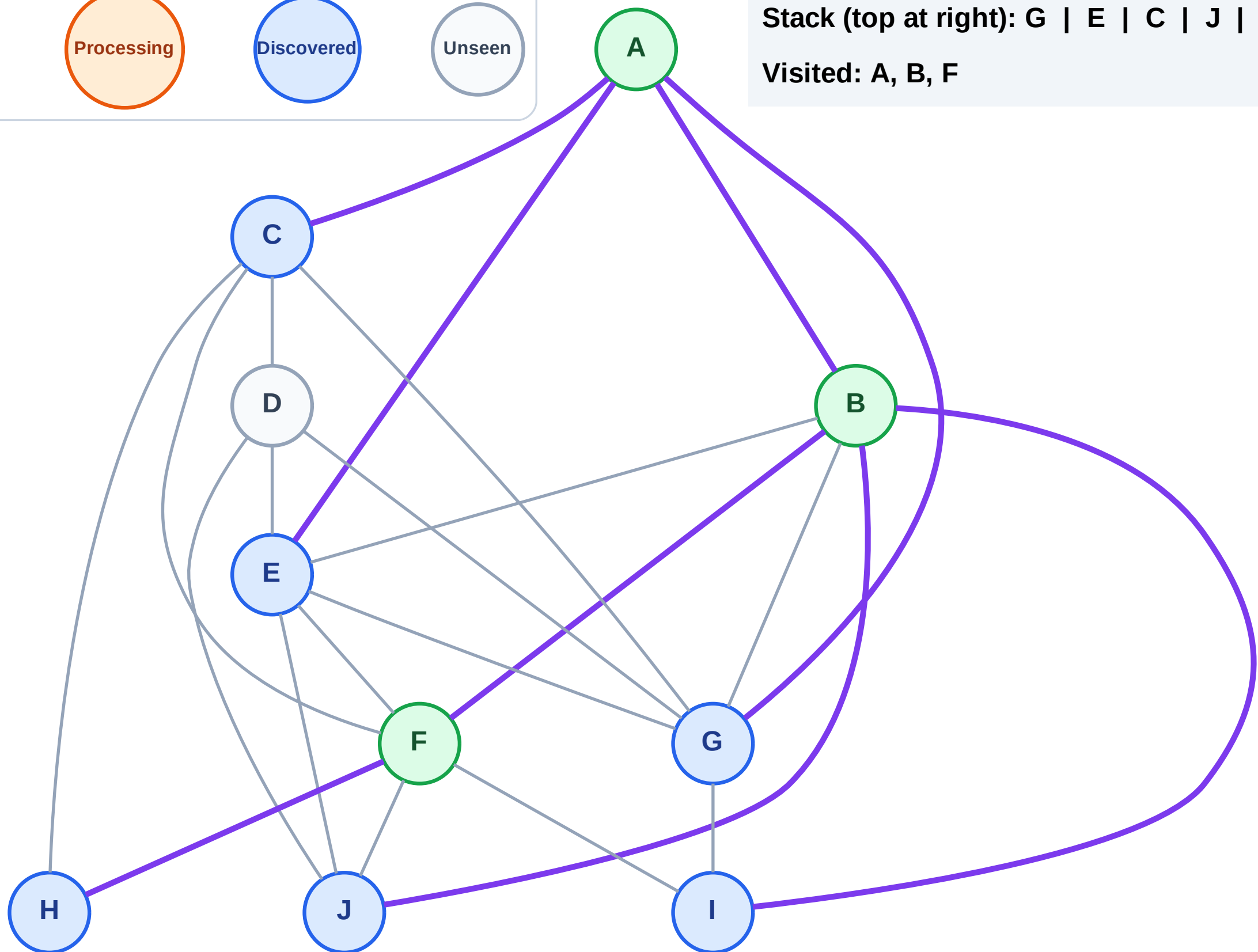
Discovered



Unseen

Stack (top at right): G | E | C | J | I | H

Visited: A, B, F



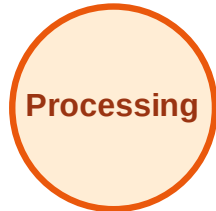
DFS Traversal

Step 8: Process node H

Legend



Complete



Processing



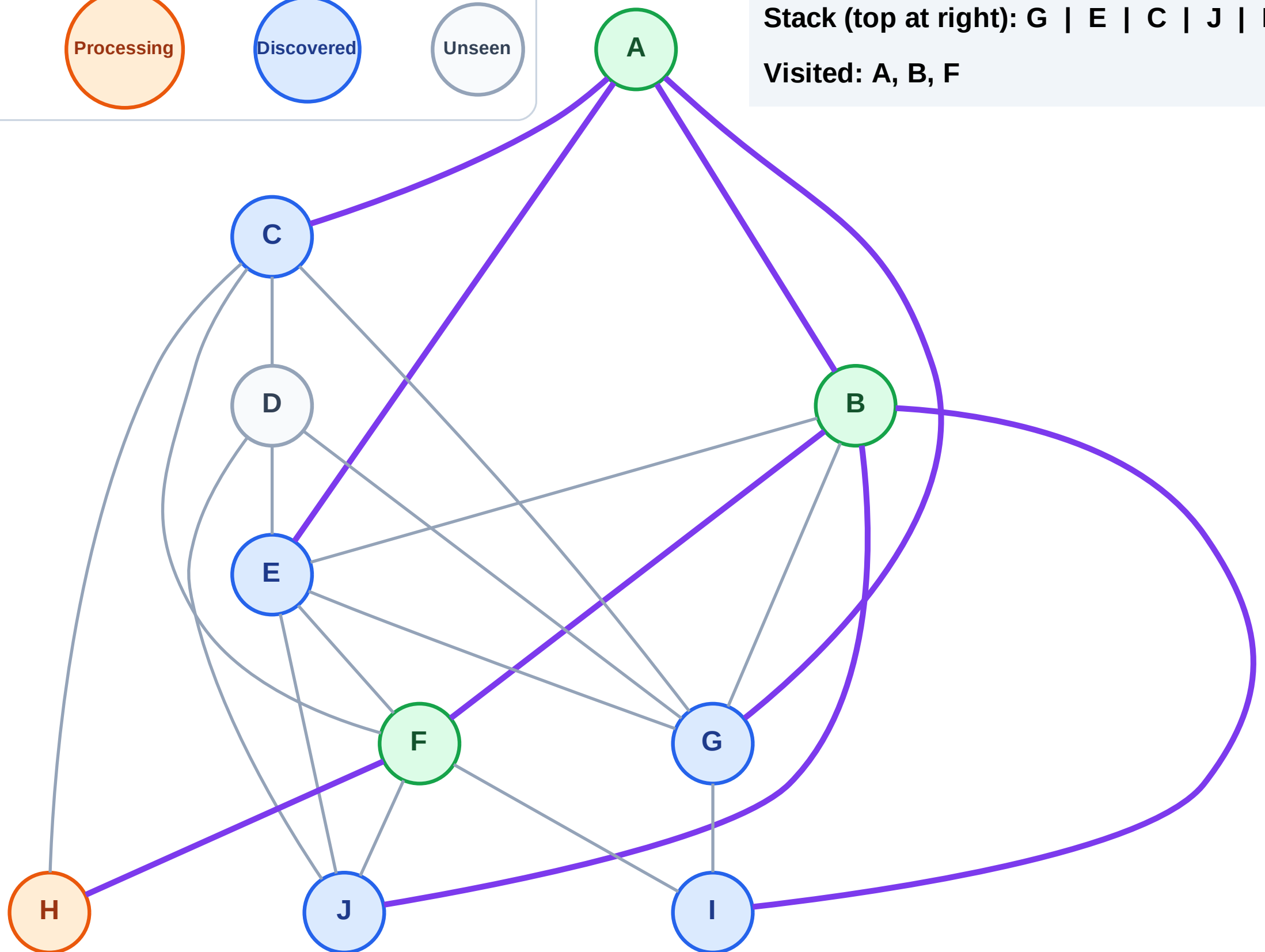
Discovered



Unseen

Stack (top at right): G | E | C | J | I

Visited: A, B, F

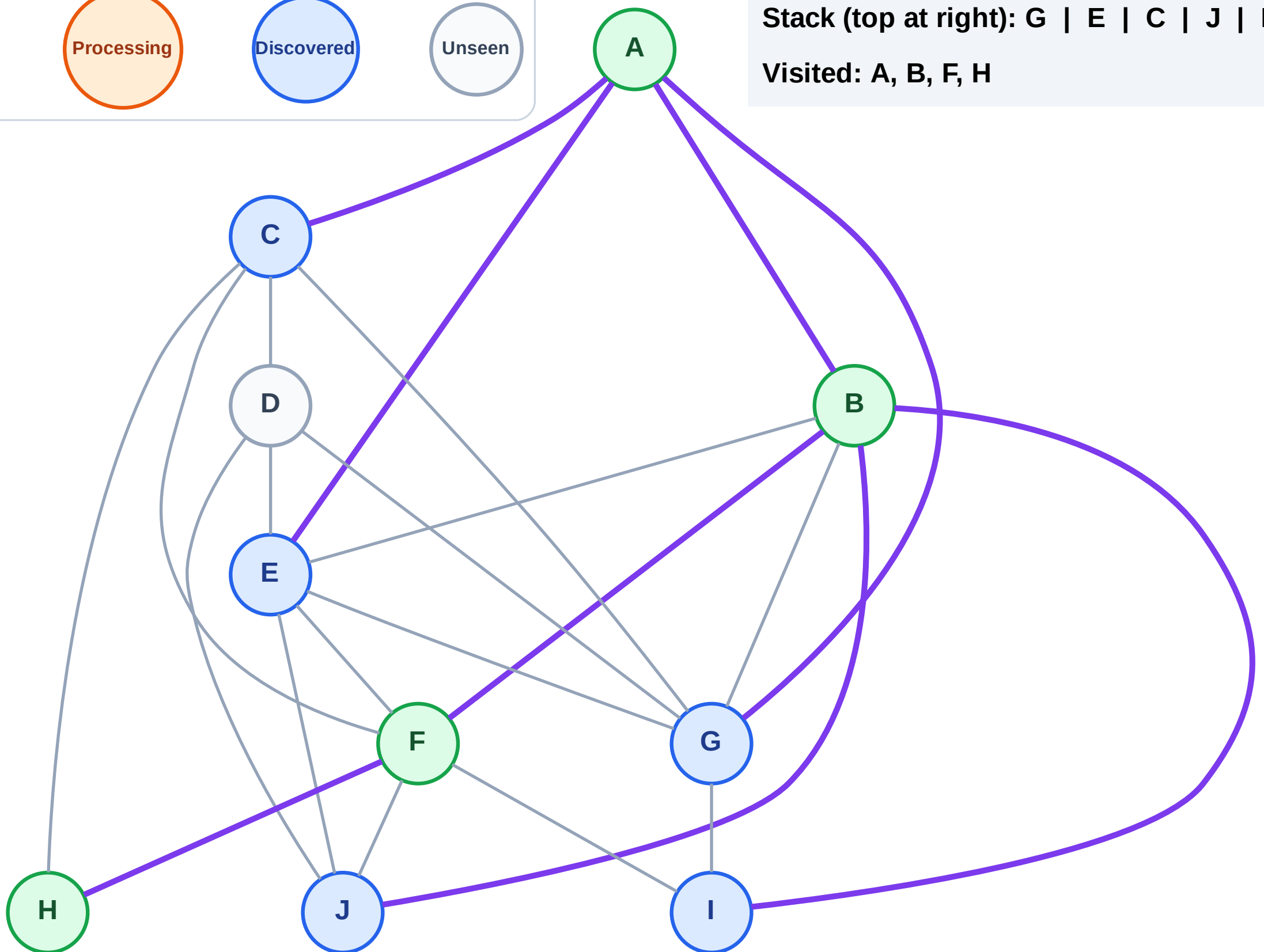


Step 9: No new neighbors from H

Legend



Visited: A, B, F, H



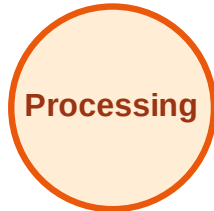
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



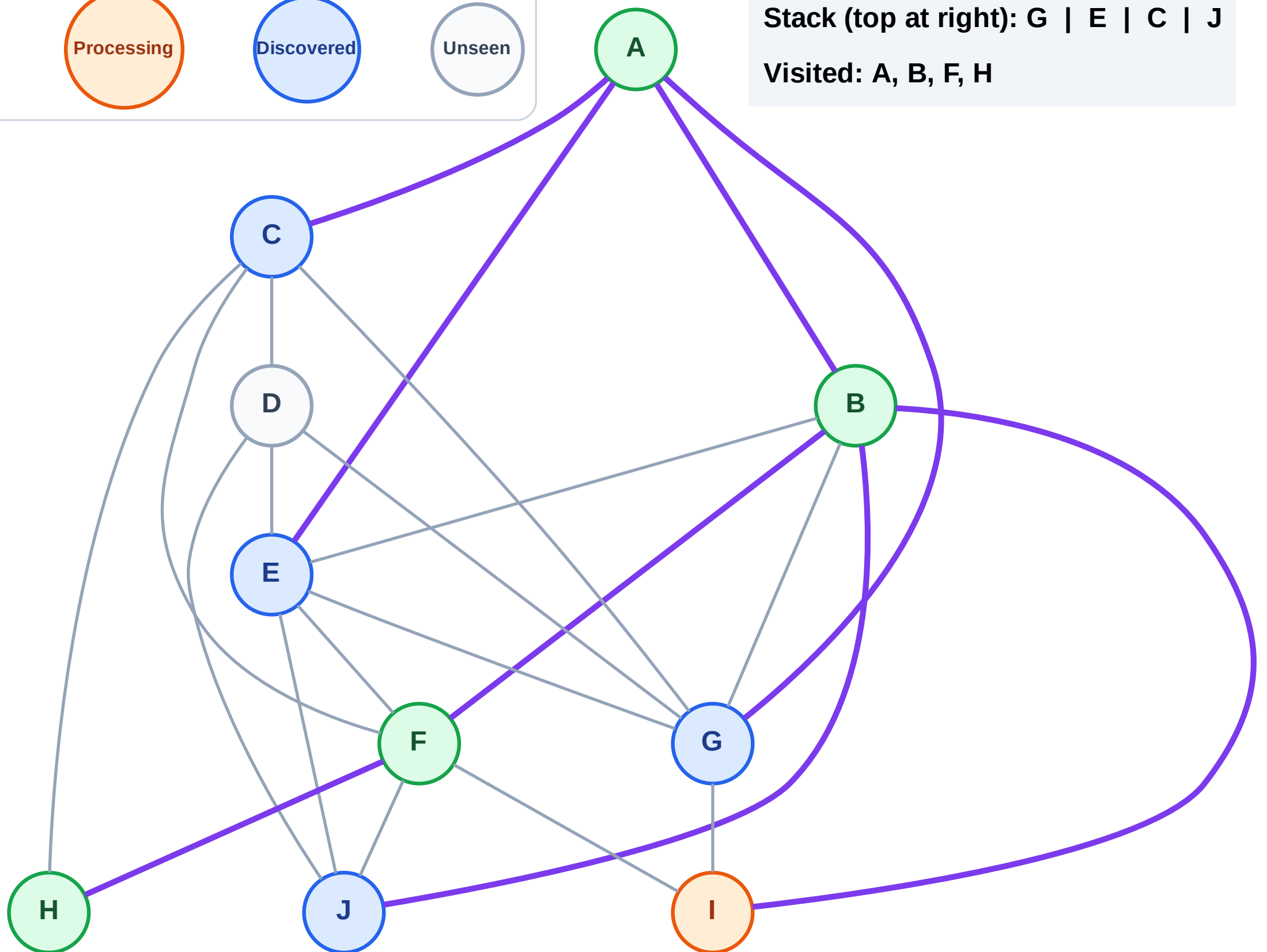
Discovered



Unseen

Stack (top at right): G | E | C | J

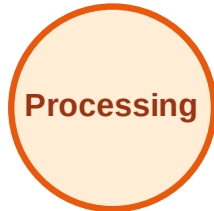
Visited: A, B, F, H



DFS Traversal

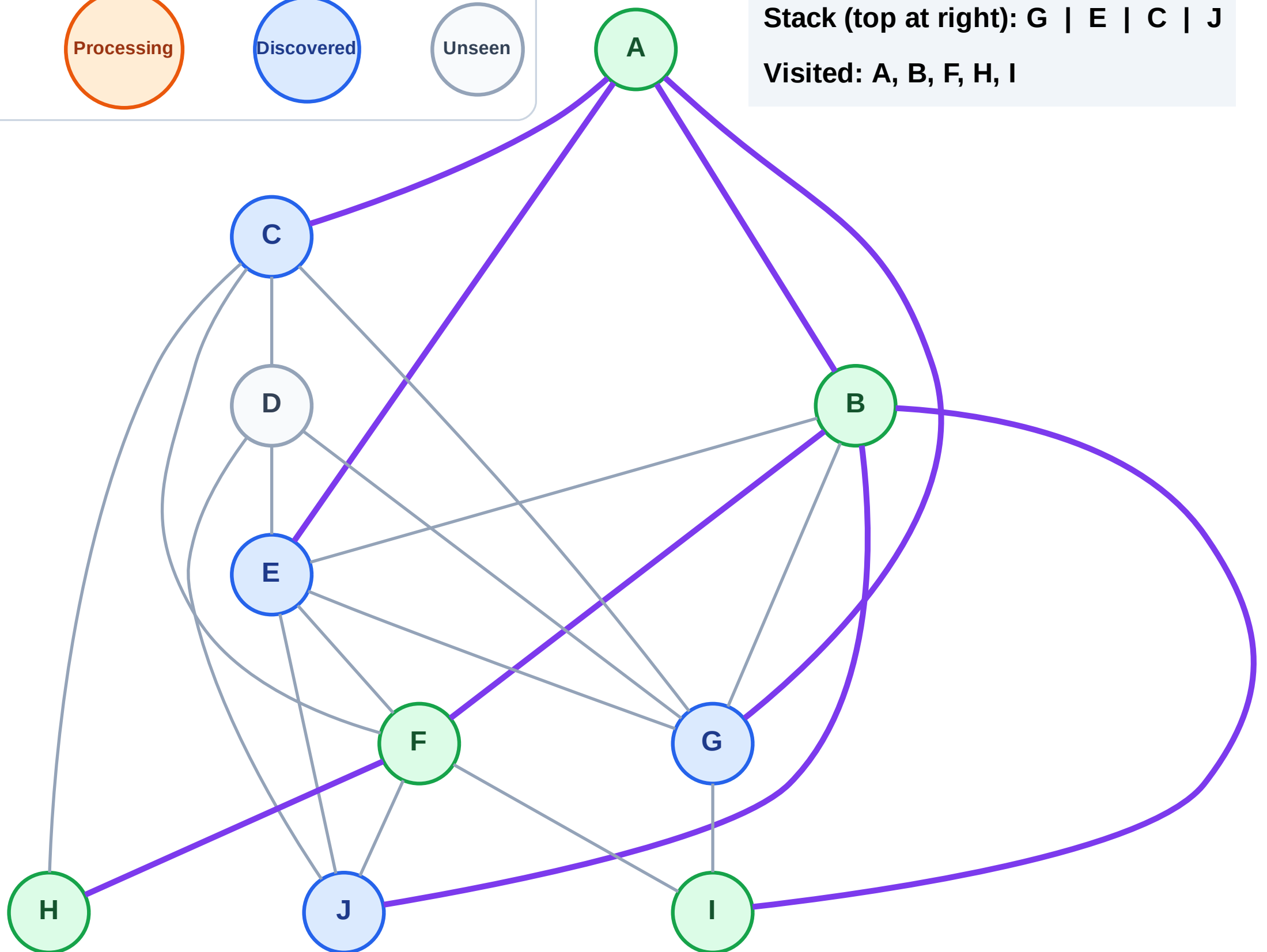
Step 11: No new neighbors from I

Legend



Stack (top at right): G | E | C | J

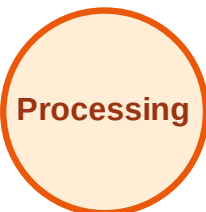
Visited: A, B, F, H, I



DFS Traversal

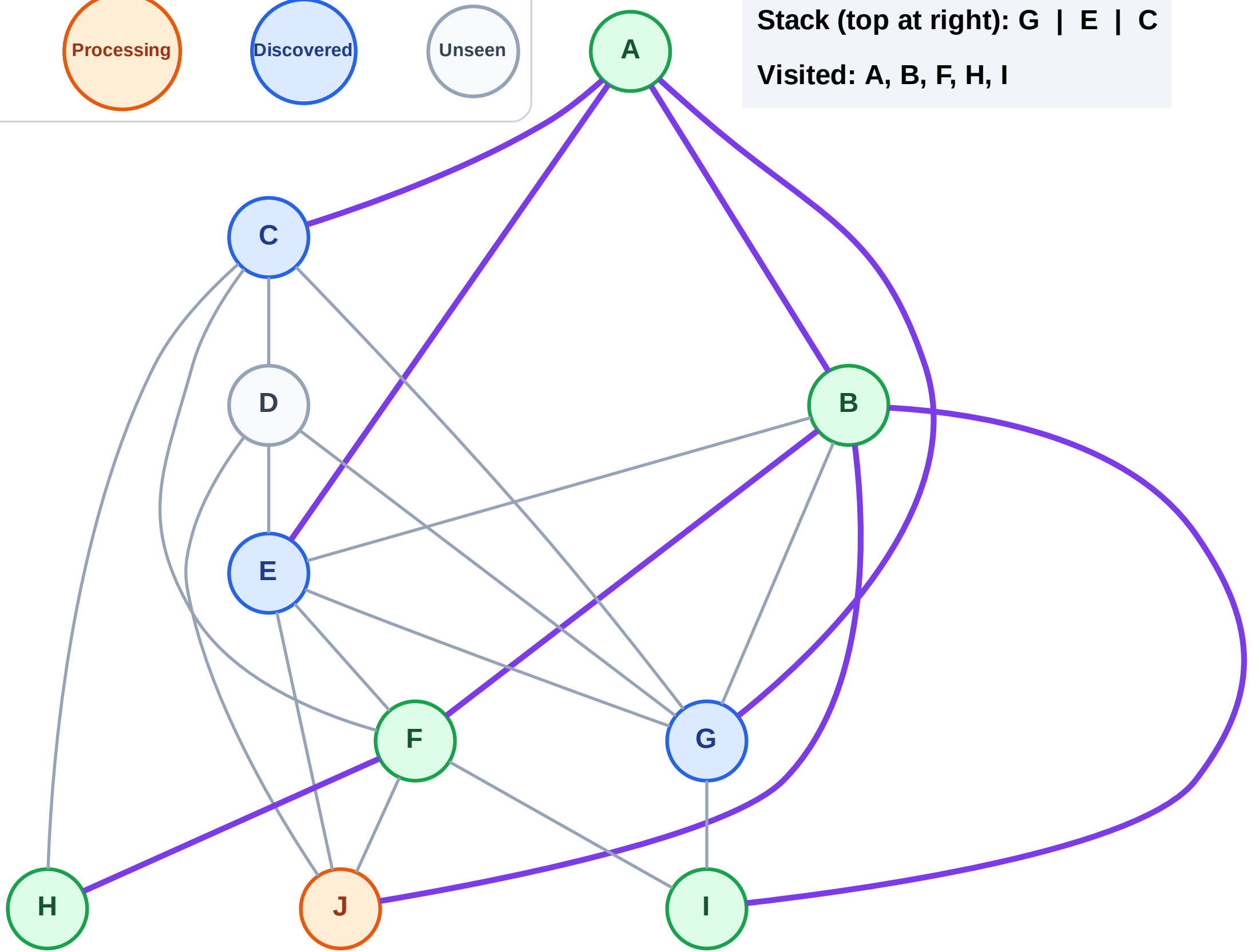
Step 12: Process node J

Legend



Stack (top at right): G | E | C

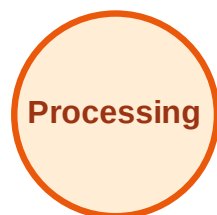
Visited: A, B, F, H, I



DFS Traversal

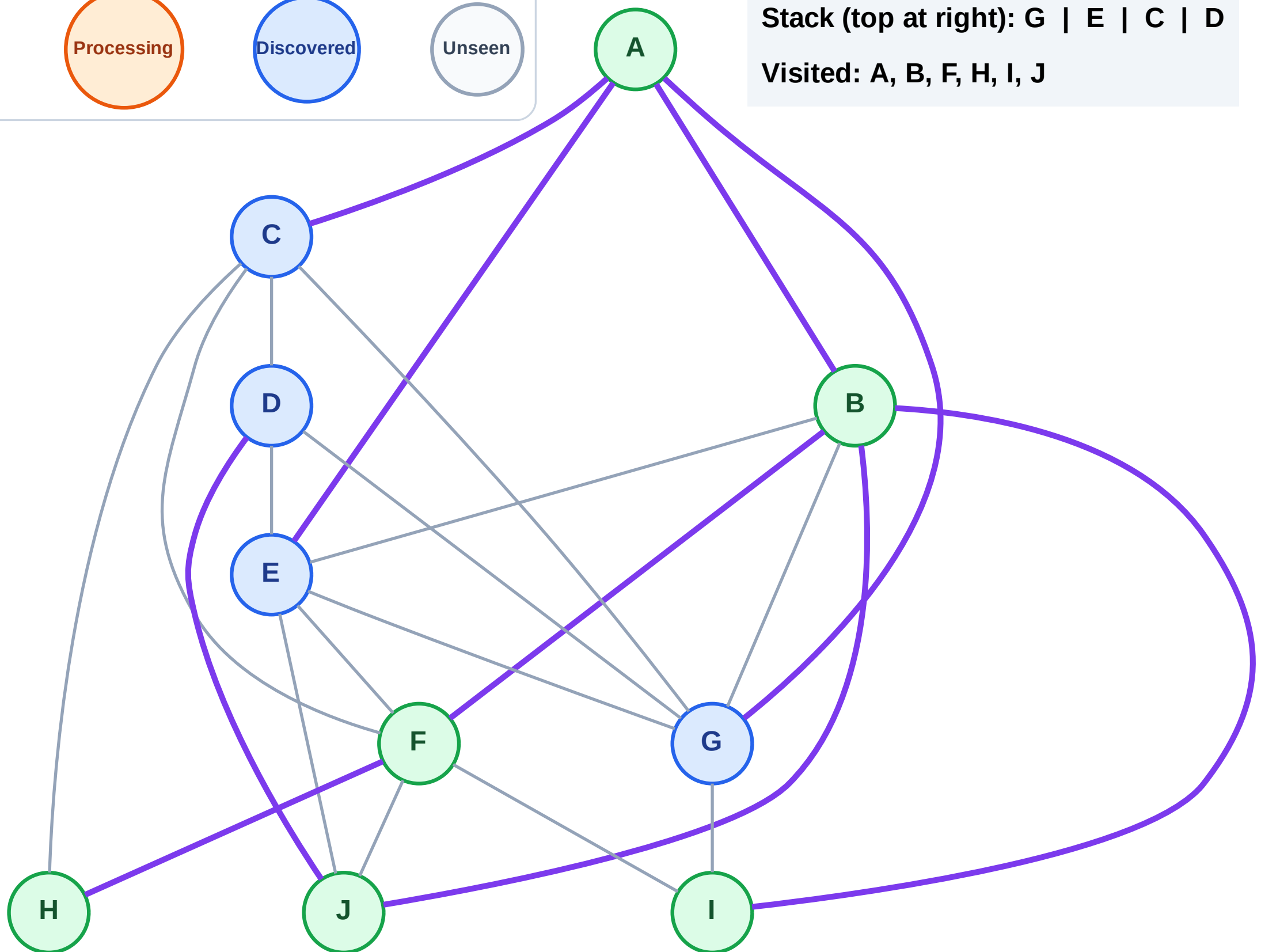
Step 13: Discover D from J

Legend



Stack (top at right): G | E | C | D

Visited: A, B, F, H, I, J



DFS Traversal

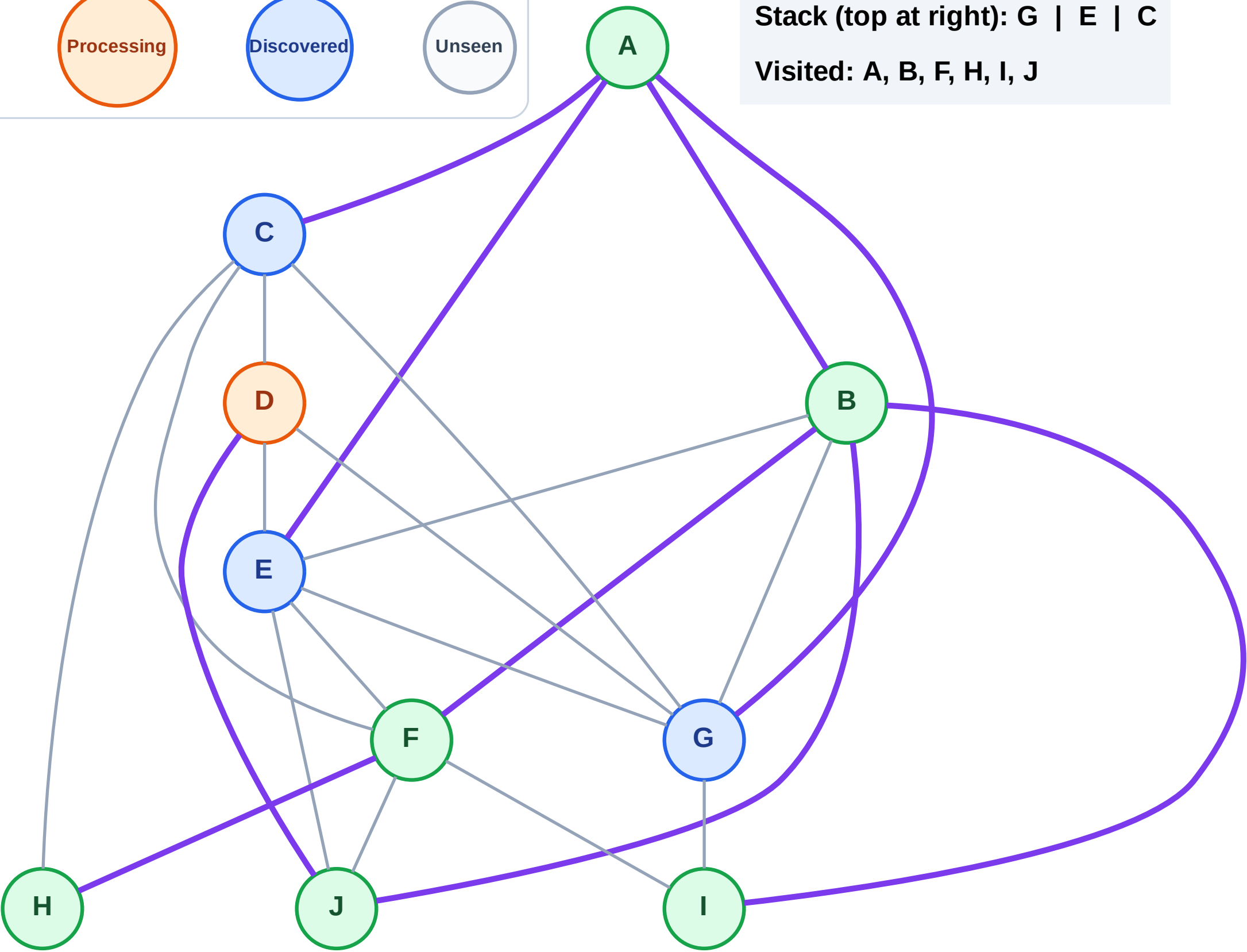
Step 14: Process node D

Legend



Stack (top at right): G | E | C

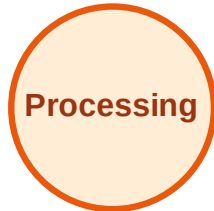
Visited: A, B, F, H, I, J



DFS Traversal

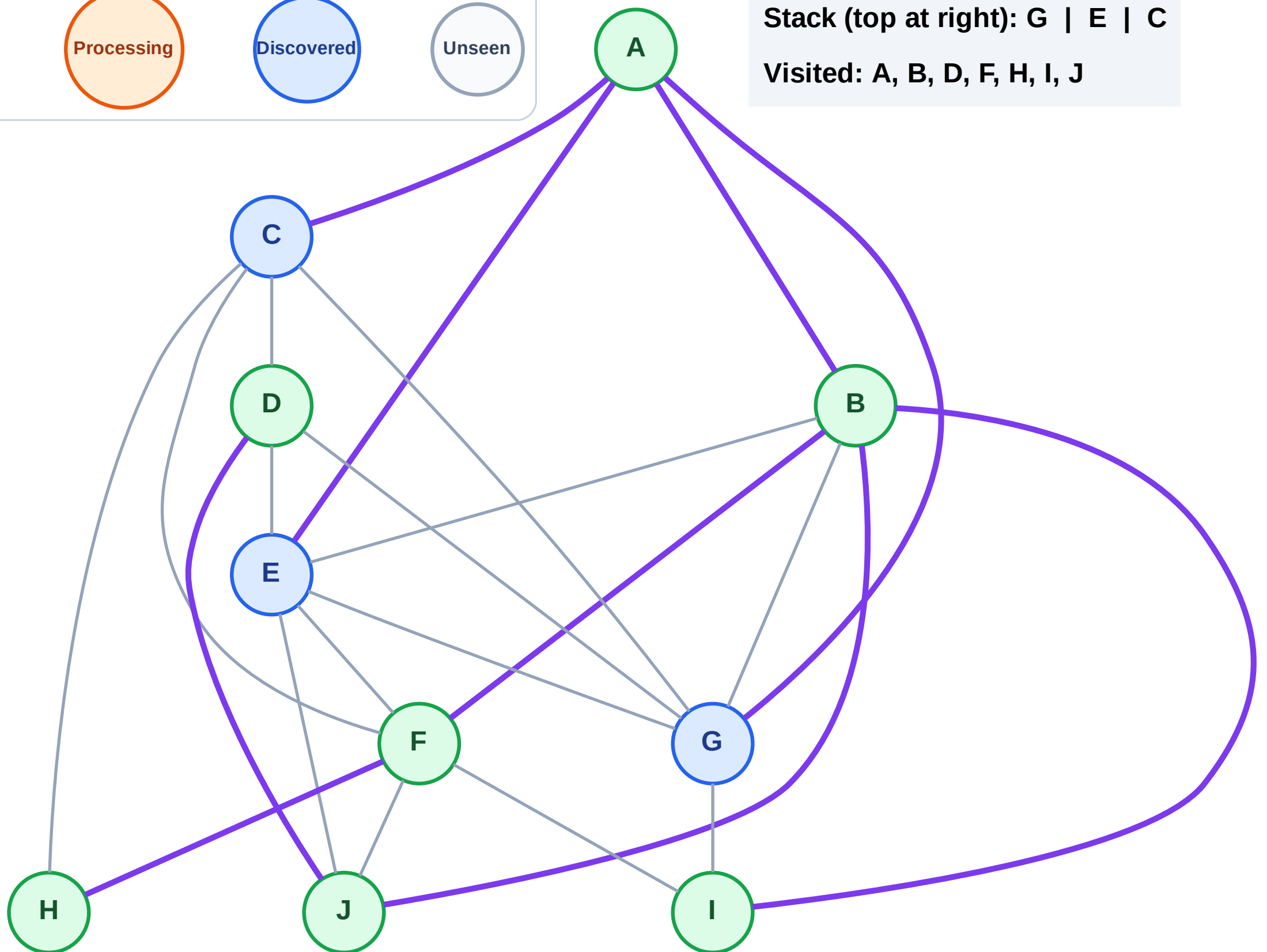
Step 15: No new neighbors from D

Legend



Stack (top at right): G | E | C

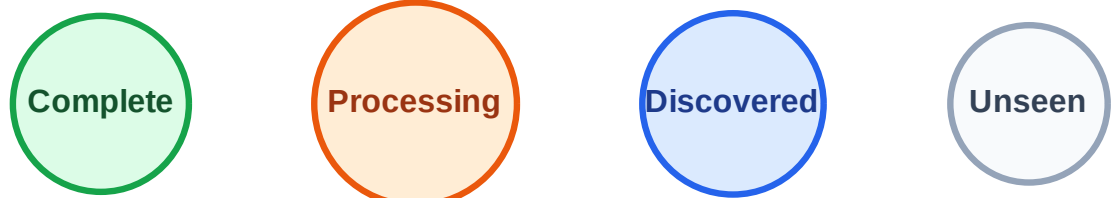
Visited: A, B, D, F, H, I, J



Step 16: Process node C

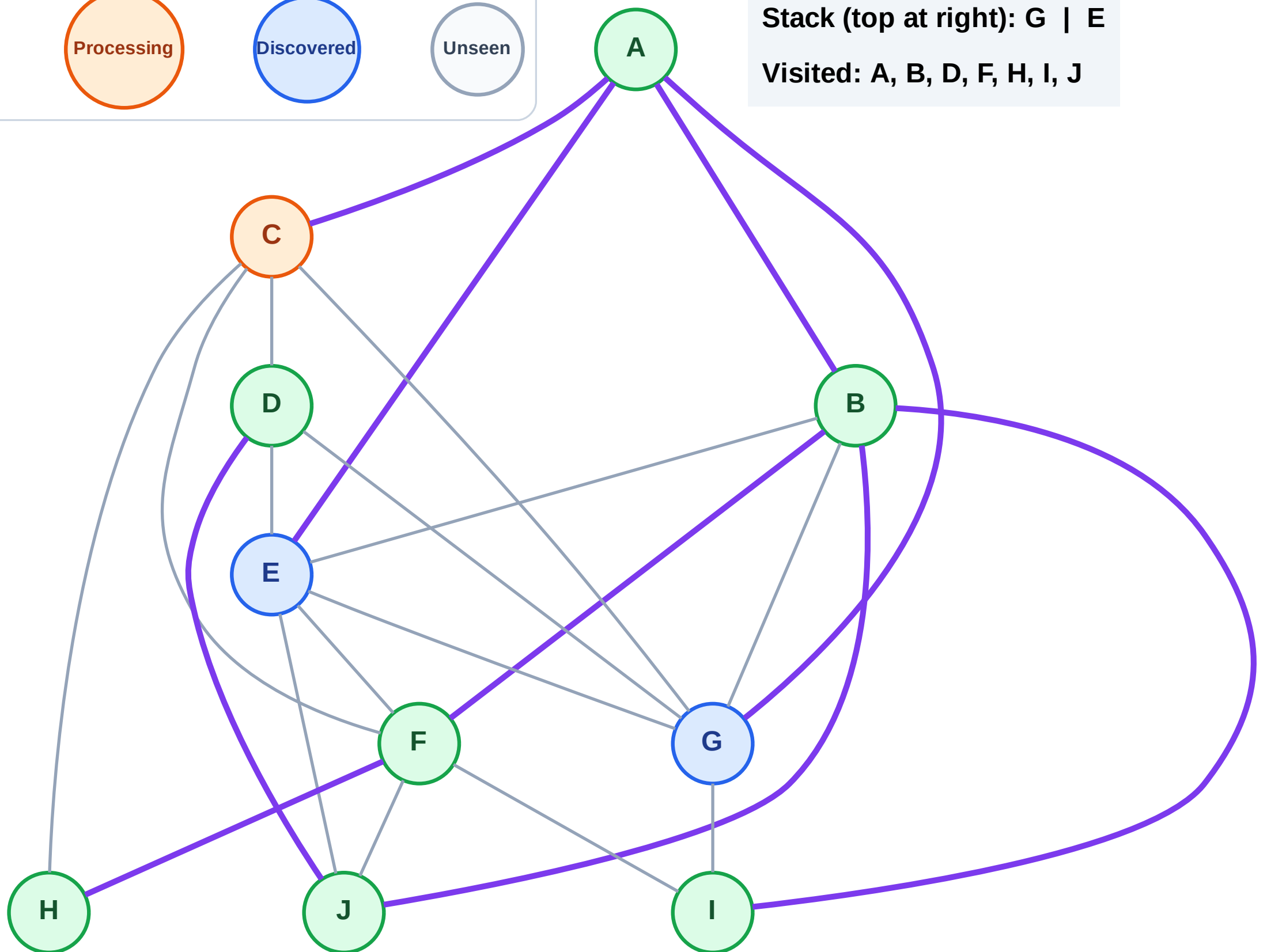
Step 16: Process node C

Legend



Stack (top at right): G | E

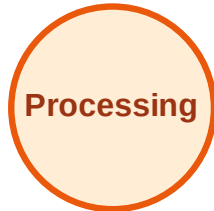
Visited: A, B, D, F, H, I, J



DFS Traversal

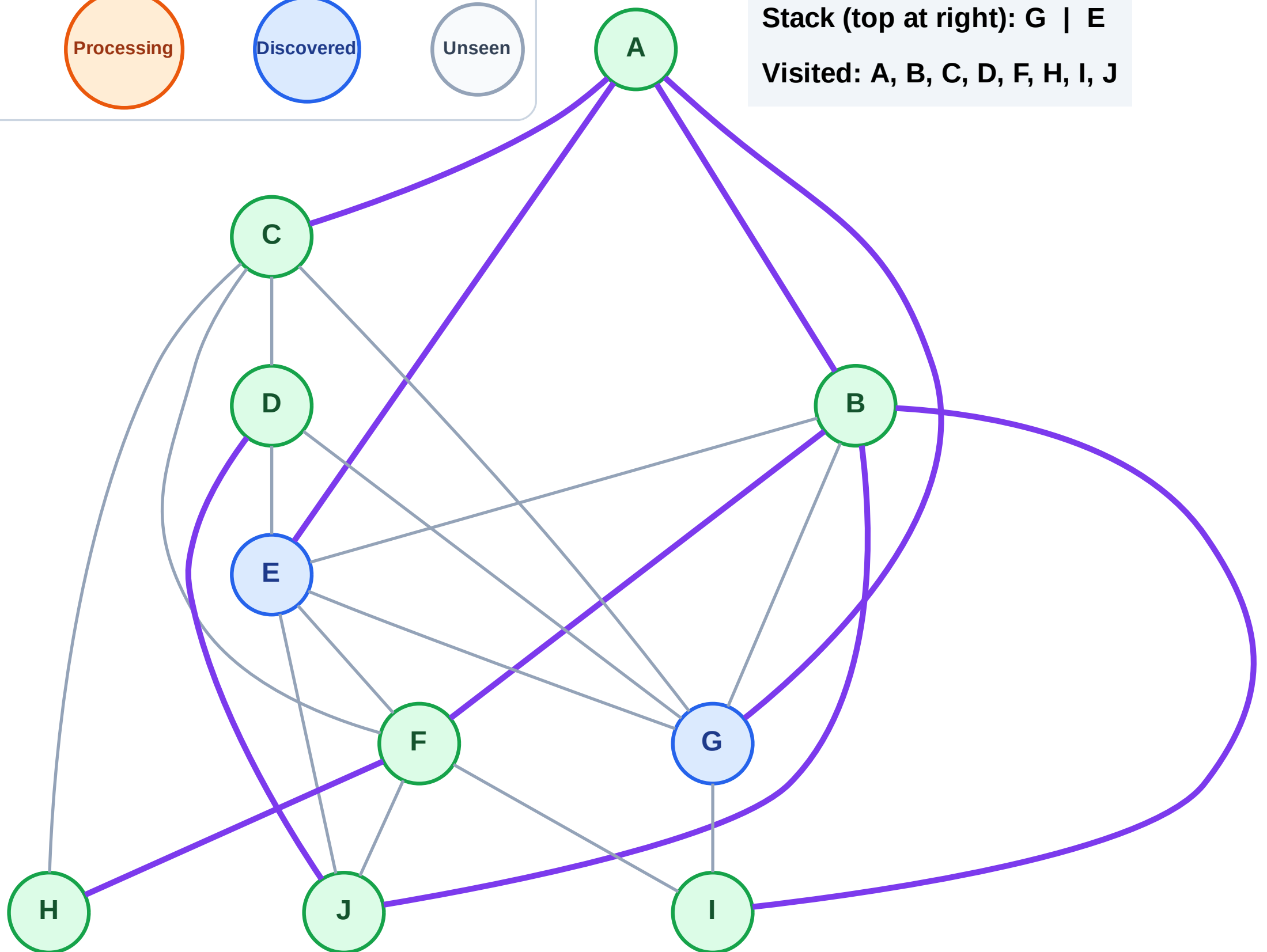
Step 17: No new neighbors from C

Legend



Stack (top at right): G | E

Visited: A, B, C, D, F, H, I, J



DFS Traversal

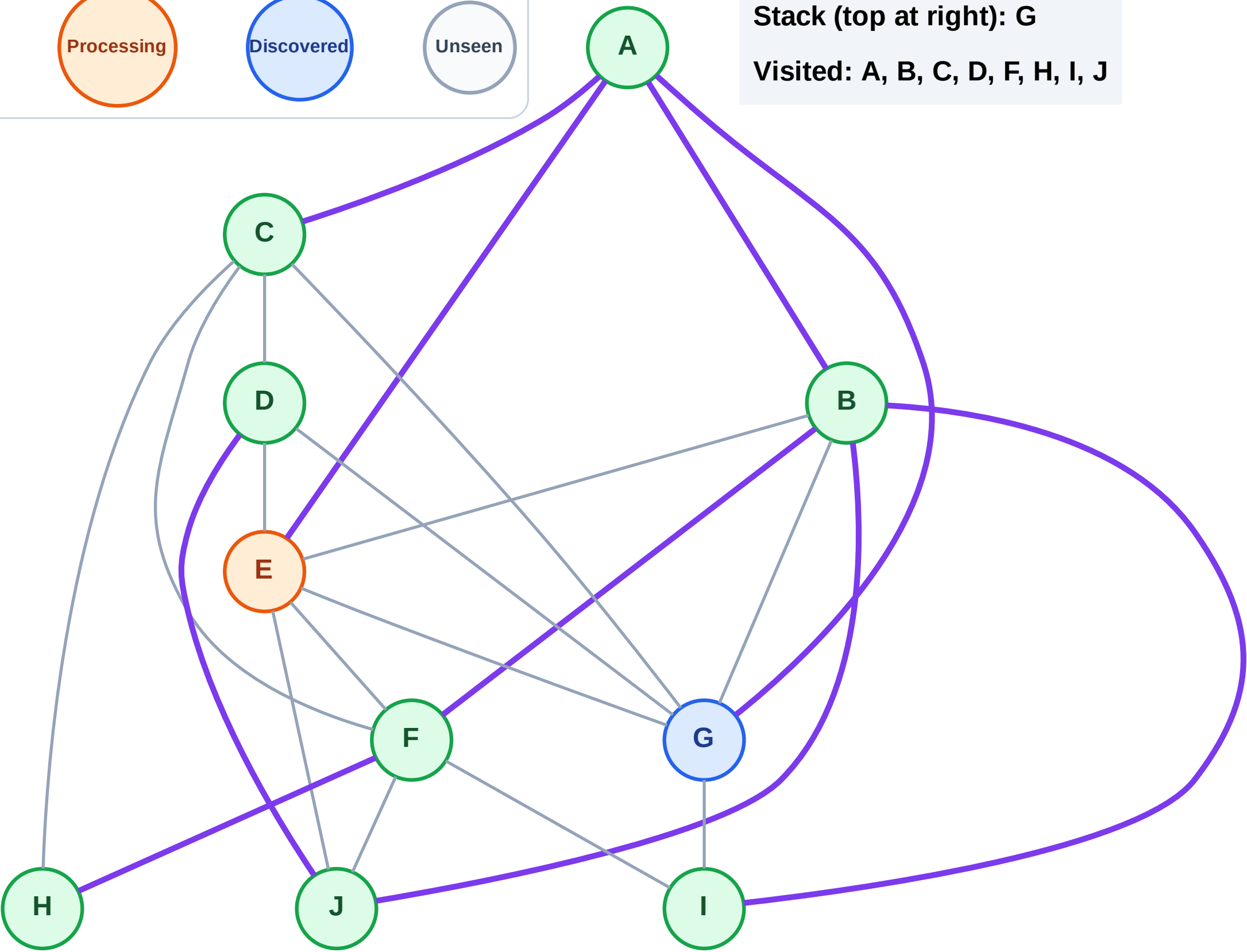
Step 18: Process node E

Legend



Stack (top at right): G

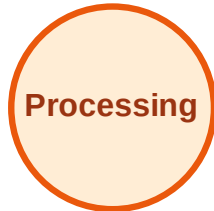
Visited: A, B, C, D, F, H, I, J



DFS Traversal

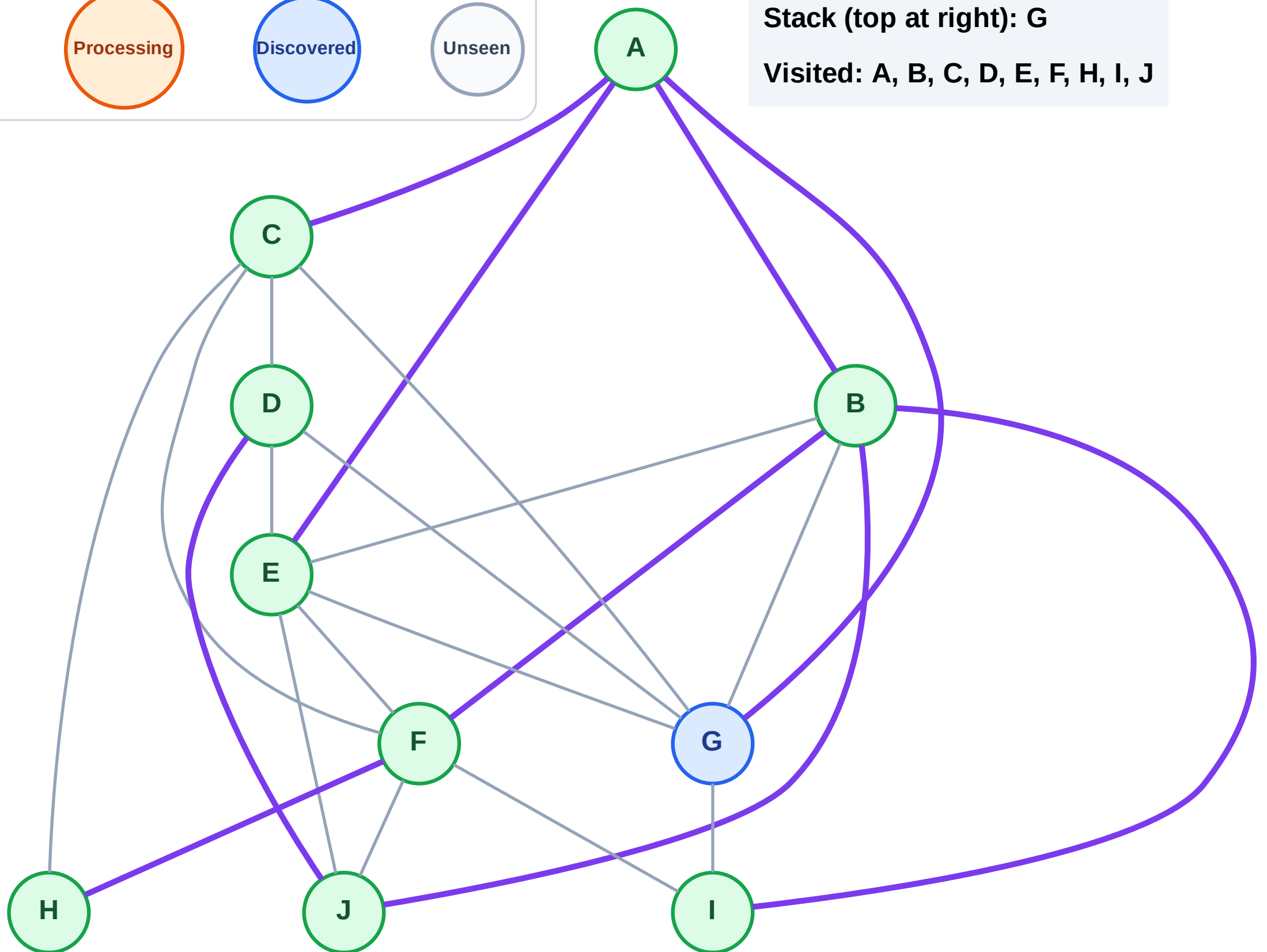
Step 19: No new neighbors from E

Legend



Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

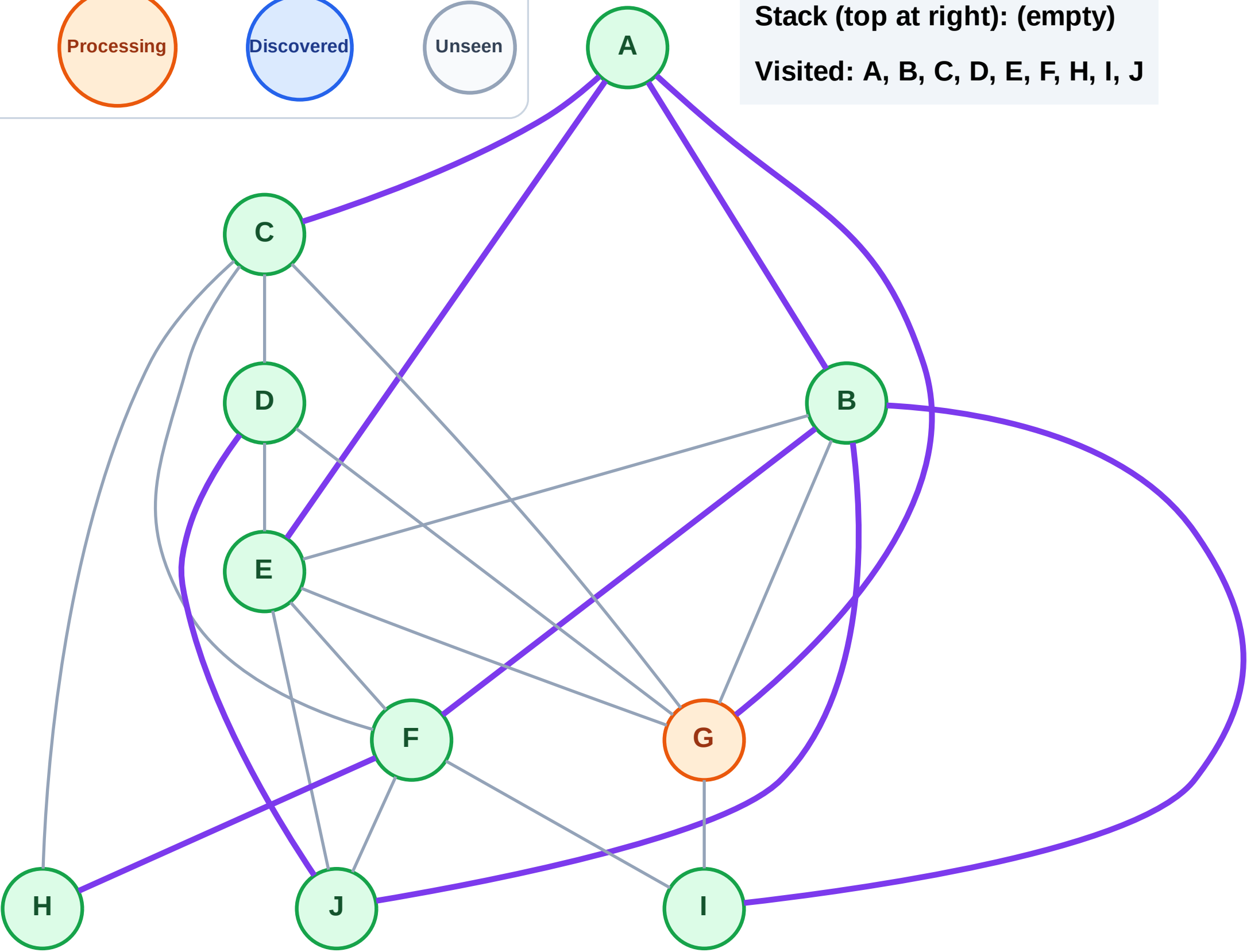
Step 20: Process node G

Legend



Stack (top at right): (empty)

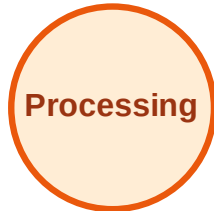
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

